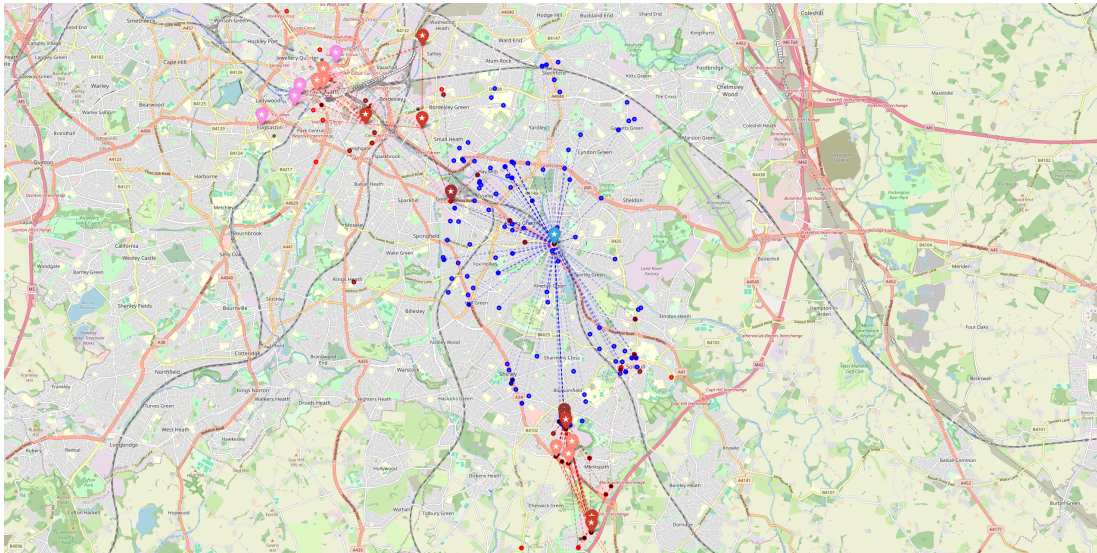


Empirical strategies to estimating spatial firm productivity spillovers using UK microdata



Firms are spatially interconnected. But are they interconnected when it comes to productivity?

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DECLARATION

I, Aloysius Lip, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

I declare the use of GenAI tools in an assistive role (UCL Category 2) for this submission. The model Gemini 3.10 Pro (Google DeepMind, 2026) was used for the limited roles of: supporting the particular process of reviewing and refactoring code (data analysis), and as a support tutor (economic model and empirical approach). The model GitHub Copilot (Microsoft, 2026) was used for the limited roles of: supporting the particular process of reviewing and refactoring code.

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Date: 2nd September, 2026

CLASSIFICATION

This piece of research is primarily:

- ☒ an empirical/econometric study
- ☐ the examination of a theoretical problem
- ☐ a critical analysis of a policy issue
- ☐ an analytical survey of empirical and/or theoretical literature

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TABLE OF CONTENTS

1	Introduction	1
1.1	Productivity in the United Kingdom	3
2	Literature	4
2.1	Agglomeration and the productivity spillover	4
2.2	Peer effects econometrics	5
3	Data analysis	9
3.1	Fame	9
3.2	Deriving panel aggregates	10
3.3	Financial and location data limitations	12
3.4	Location group classification	14
3.5	Location distance calculations	17
3.6	Higher-order $\mathbf{W}$ transformations	20
4	Static models	22
4.1	Static panel identification	24
4.2	Linear-in-means group model	28
4.3	Static distance decay	33
5	Dynamic models	36
5.1	Dynamic panel identification	37
5.2	Exploring temporal lag structures	38
5.3	Higher-order instruments	42
5.4	Dynamic empirical results	44
6	Discussion and extensions	45
	Bibliography	47
	Appendix	52
A	Fame data computational methods	52
A.1	Data cleaning procedure	52
B	Extensions	56
B.1	Selection of parameters	56
B.2	Improving estimates by granularity	56
B.3	Network fixed effects	57
B.4	Alternative location data	59
B.5	Vector transformations procedure	60
B.6	$\mathbf{W}$ transformations summary statistics	62
C	Static groups: additional results	64
D	Static distance-decay: additional results	66
E	Data cuts: estimations with subsets	67

LIST OF FIGURES

I	Illustration of group matrix $\mathbf{W}_g$	15
II	Illustration of multiple, non-overlapping groups	16
III	Histogram of distances between any target-firm pairing	18
IV	Illustration of distance-weighted network matrix $\mathbf{W}_d$	19
V	Illustration of overlapping groups from a target firm to its whole network through successive peers	21
VI	Panel observation availability by year	54
VII	Distribution of years of data available per firm	54

LIST OF TABLES

I	Spatial econometric identification rules for weighting matrix $\mathbf{W}$	8
II	Capital-output ratios in FAME corpus of firms	10
III	Benchmark 2-factor TFP estimation	11
IV	Weighting matrices i, j -relation properties	14
V	Summary statistics across location groups	16
VI	LMM: 1 peer regressor with group weights, error orthogonality model vs instrumented	30
VII	Static panel IV using continuous distance-decay $\mathbf{W}_d$	34
VIII	Dynamic panel: baseline instrumental specification	41
IX	Dynamic panel: deeper instrumental lags	43
X	Summary statistics for $\mathbf{W}_d\mathbf{x}$ and high-order transformations . .	62
XI	Summary statistics for $\mathbf{W}_g\mathbf{x}$ transformations and high-order variants	63
XII	First-stage 2SLS (panel OLS) results	64
XIII	Static panel IV, 3 orthogonal groups	65
XIV	First-stage 2SLS (panel OLS) results	66
XV	Static panel IV using continuous distance-decay $\mathbf{W}_d$, zero dis- tances excluded	67
XVI	Spillover estimation for 4 select innovation clusters	68
XVII	Heterogenous spillover effects among service-sector firms in UK cities	68
XVIII	Heterogenous spillover effects among service-sector firms in UK towns	69

ABSTRACT

This paper offers empirical strategies across static and dynamic models to estimate the productivity spillover occurring between colocated firms in the United Kingdom. The productivity spillover, a spatial externality, underpins well-established economic phenomena around the development of urban clusters and regional economic policy.

I examine approaches to estimate the spillover using a minimally-restrictive, firm-to-firm model. I develop a 3-factor production model of Total Factor Productivity (TFP), capital, and labour, with TFP in turn decomposed to spillovers of those same three factors from peers. I apply econometric techniques for valid identification in peer effect group and network settings to UK mass microdata from Fame. The key to valid identification rests in internal instruments from spatial *exogenous* effects in a static model, and from deep lags in a dynamic model: this overcomes Manski (1993)'s 'Reflection Problem' and the endogeneity of peer effects. When faced with big data, this requires complex transformations to calculate appropriate instruments or fixed effects; I show this is possible using modern data techniques.

The estimates offered in this paper do not conclude substantive estimates of the commonly-reported agglomeration elasticity. Instead, I focus on solving several econometric problems for more robust estimation on a targeted basis, including the application of a peer effects model to a dynamic panel IV setting and identifying cases of endogeneity with solutions. The evidence provided suggests that static models can be robust, and composite instruments have the potential to improve efficiency in a dynamic setting.

Targeted extensions building on this paper's framework include estimation of the heterogeneous strength of the spillover effects among different sets of actors and the development of network fixed effects.

1. INTRODUCTION

The strength of the spatial productivity spillover is crucial to policy and economic discussions on growth, production, and the formation of urban clusters, but it is notoriously tricky to estimate⁴. The purpose of this paper is to demonstrate that modern advances in rich microdata compilation, computing, and econometric theory and techniques mean that progress is possible in this decades-long empirical problem. Indeed, evidence is derived by stripping away restrictions and stylised patterns of firm and household behaviour that simplify the problem in empirical literature, in favour of estimation with a stylised, classical model only using features available from firm data. This provides a baseline technique for future, more sophisticated models. The analysis offers four key contributions that are substantively novel:

1. The development of data science techniques to calculate spatial transformations on rich microdata to fulfil exogenous regression specifications
2. The application of higher-order spatial lags as an internal instrument deploying a peer effects model in a firm productivity setting
3. The application of higher-order temporal lags as an instrument in a dynamic panel peer effects model, contrasting instrument performance
4. Empirical estimates of the firm-to-firm spatial productivity spillover. This is evidence that can underlie patterns of firm colocation discussed in the more sophisticated urban development literature, and provides a framework for future extensions to investigate heterogenous effects.

This analysis acknowledges a large body of work on agglomeration which elucidates the importance of urban density and development on firm productivity. In section 2, I give an overview of the substantive contributions from urban economics and New Economic Geography (NEG), which explain agglomeration driver channels and offer sophisticated approaches to modelling to household and firm dynamic location choices. No attempt is made to recreate this whole body of work. Instead, this analysis abstracts away from the urban modelling choices common to this literature. Despite being a spatial analysis, location is employed only as a feature to identify a group of spatial peers for every firm. From this mere spatial proximity, this analysis reviews approaches to generating empirical evidence on the pure productivity spillover from peers to a target firm, highlighting approaches to overcome empirical challenges on the way. If successful, evidence derived from this framework may provide *prima facie* support for the centrifugal

⁴ Krugman (1998), Fujita and Krugman (2004)

force driving urbanisation and cluster formation. This is crucial to both the descriptive analysis of economic geography and the prescription of regional policy.

Section 2 also gives an overview of the econometric literature on peer effects, from Manski (1993)’s ‘Reflection Problem’ to evaluating instrumental validity in a more complex graph and network setting in Bramoullé et al. (2009). This analysis applies that econometric theory to the productivity spillover setting.

In sections 3 and 4, I discuss the handling of data from Fame, a compilation of publicly-available, balance-sheet data from UK companies designed for comparative research. The cleaned Fame dataset provides more than 150,000 firms’ data and 1.1 million panel observations. Section 3 also highlights data science techniques developed for this paper to undertake complex matrix transformations of higher-order spatial lags of an $(n \times n)$ interactions matrix necessary to compute valid instruments and eliminate endogeneity. I provide a detailed guide to reproducing methods in Appendix A, with all code available publicly⁵.

In section 5, I give the econometric analysis of a static model with spatial peer effects, conceptualisable as ‘spatial lags’ of the classical regressors in a 3-factor production function. This contrasts the merits of different weighting approaches to analysing peer groups and the use of spatial lags as valid instruments to the peer effects *endogenous* regressor. I estimate both ‘groups’ models, where the strength of a productivity spillover is the average of nearby neighbours, and ‘networks’ models, which applies a distance-decayed individualised weight for every target-peer firm pair.

In section 6, I expand the use of internal instruments to temporal lags. This builds on modern ‘System GMM’ methods from Blundell and Bond (1998). I discuss advantages of relaxing strict assumptions to ‘weak exogeneity’ and evaluate the predetermined nature of classical production factors in a dynamic setting. These models offer minimal evidence of a dynamic autoregressor coefficient, validating the use of static models which restrict effects to the contemporaneous period.

Throughout this analysis, I prioritise extensions which expose the econometric properties of the model, establish a grounding for future analysis. I propose that this baseline establishing an investigation method, with credible contributions to the data science, theory, and econometric techniques of issue are sufficient for the scope of this paper. Policy-relevant extensions are discussed in Appendix B.1, which I leave it for further analysis to execute. These have the potential to investigate the policy-meaningful questions I discuss in this introduction.

⁵ <https://github.com/alloysiuslip/ucl-diss-ECON0079>

1.1. PRODUCTIVITY IN THE UNITED KINGDOM

The United Kingdom was a regional economic powerhouse through the Industrial Revolution. From the concentration of textiles in Manchester to car manufacturing in the West Midlands, as the natural geographic distribution that lent itself to heavy industries, so did national interest in regional economic policy. Regional focus declined strongly in the UK under the 1980s Thatcher government and was largely maintained at lows throughout the 1990s-2000s New Labour period, where rapid economic growth known as ‘The Great Moderation’ satiated impetus for change.

The consensus was broken in 2017, with the seeds of a regional-industrial strategy in 2017, emerging under the Conservative May government. Under the background of a UK-EU Referendum result (“Brexit”) which emphasised the sentiment of disaffection among ‘left-behind’ places across the UK which did not see the benefits of national aggregates of economic growth primarily concentrated in the UK’s capital, London. The national government launched ‘Local Industrial Strategies’ which sought to identify action to "boost productivity, earning power, and competitiveness" on a regional basis⁶. However, fewer of these strategies were fully developed and implemented by 2021.

A revival of interest in regional, pro-second city policy has also taken place in the policies of the 2024 and 2026 governments in the United Kingdom. In 2026, the newly appointed Burnham government made the pledge for “good growth in every postcode”⁷ with immediate speculation and interpretation from economic commentators⁸ on the details for the direction of regional and urban policy in the United Kingdom in the absence of further details.

It is well understood through the literature that follows that high productivity tends to follow urban clusters, and specialisation and devolution can be an engine for economic growth if deployed appropriately. However, the evidence on that appropriate manner is muddled at best, with most political and economic insight limited to observing and recounting trends of recent cluster development. There have been modern success stories in the United Kingdom, such as the sudden and successful rise of Canary Wharf in London as a financial hub, the emerging life sciences cluster in Cambridge and the ‘Golden Triangle’, and creative industries such as video-game development in Edinburgh and the North of England. Nevertheless, with the UK government pushing to go further and deploy substantial fiscal tools as a means of regional productivity shocks⁹, the crucial policy question

⁶ Saywah Mahmood and Georgina Hutton and Lewis Pickett (2025)

⁷ GOV.UK (2026)

⁸ Matthew Gill (2026), Muthu De Silva (2026)

⁹ HM Treasury (2026)

remains on the endogenous sustainability of a spatially-connected system. Can the rise in productivity associated with clusters be attributed internally to the cluster itself, or is the result of repeated, other external factors? Guidance on this is crucial for effective policy. This analysis seeks to establish a baseline from which that evidence can be generated.

2. LITERATURE

2.1. AGGLOMERATION AND THE PRODUCTIVITY SPILLOVER

Agglomeration is the colocation of firms in spatial regions for the mutual benefit of a productivity spillover. It is the imperative force behind the development of cities (described illustriously in Goldin and Lee-Devlin (2023)) and fundamental to considering alternative dimensions to the static 2-factor production model underlying most economics of firm production that mostly intuitively represents the pattern of growth and clustering seen in modern economies. Public attention was drawn to the phenomenon even in the heyday of the development of classical economics. Marshall (1920) outlined a series of channels pertinent to increasing urbanisation around industrial clusters that characterised the late 19th Century across Europe. Agglomeration was definitively described by Krugman (1991), an empirical investigation into the clustering of manufacturing firms using a ‘core-periphery’ model of urban development. The paper, and the authors’ subsequent work of the ‘New Economic Geography’ field (Krugman (1998); Fujita and Krugman (2004); Koo and Lall (2007)), was nuanced in its characterisation of ‘centripetal’ forces, grounded in Marshall (1920)’s taxonomy, accelerating towards a basin of attraction urban area until they are outweighed by the ‘centrifugal’ forces of a developed city: immobile factors, competition for resources, and congestion.

The movement spurred other considerable advances in spatial economics, which looked to estimate the raw underlying spatial spillover effect, supporting ‘New Economic Geography’ contributions which tend to assume or abstract away from the spillover estimate itself. Several contributions outlined in Beeson (1991) robustly estimate the contribution of increased city size to productivity levels; later work takes this on a more granular level, developing estimates of the contribution of indices of spatial and industry density for any given urban area (Glaeser et al. (1992); Ellison et al. (2010)).

Density-based estimates of the agglomeration ‘centripetal’ force provide a different interpretation to spillovers from firm-to-firm, both to implications on economic policy and their own confounding issues. Contributions at the turn of the 20th century offered some firm-level estimates, but tended to continue to make significant abstractions, such as imposing the equilibrium model of firm

movement across 2 abstract cities or the placement of firms all on a 1-dimensional line Fujita (1982); Ciccone and Hall (1996); Desmet and Rossi-Hansberg (2010). Nonetheless, these offer benchmarks to empirical estimates derived and helpful insights on extending the classical production functions and deploying distance-decay measures to simulate the spatial characteristic of knowledge decay or firm-firm interaction barriers, such as in Lucas and Rossi-Hansberg (2002), which I emulate in this paper.

Estimates are summarised in Graham and Gibbons (2018), commissioned for the UK Department for Transport. They evaluate a mean across 47 international empirical studies around 5%, positively signed, though noting the wide variance in estimates. They additionally lay out a step-by-step framework to estimating firm-to-firm productivity spillovers through a production function approach. Both group and distance-decay measures are discussed. Additionally, an overview of the measurable data and econometric options to carrying out an empirical investigation are comprehensively provided. This analysis implements their suggestion to employ a dynamic panel IV, alongside other spatial IV techniques as an instrumentation strategy. Recent papers utilising these econometric techniques, coupled with exploiting modern, public, mass data-collection have offered further insightful estimates on the firm productivity spillover in specific contexts. Bloom et al. (2013) uses instruments on innovation to isolate ‘centripetal’ and ‘centrifugal’ forces. Gruber et al. (2023) instruments place-based cost shocks to evaluate the pattern of colocation and productivity of scientists. There are few estimates that seek to take a UK approach to estimating this spillover factor on a firm-to-firm level.

Graham and Gibbons (2018) additionally substantively review the econometric pitfalls of estimation that challenge a spatial econometric investigation at the firm-level. They highlight the critical issue addressed in this paper on the simultaneous determination of the dependent variable and the causal productivity spillover, which causes estimates with a reduced-form TFP residual dependent variable to be “biased and inconsistent”¹⁰ This paper’s advances on this literature are to move away from aggregated density measures to a firm-to-firm spillover, address endogeneity via a peer effects model.

2.2. PEER EFFECTS ECONOMETRICS

The Manski (1993) ‘Linear-in-Means’ model provides a baseline for the econometrics of peer effects. It provides helpful structure to examining joint behaviour of

¹⁰ Graham and Gibbons (2018), p.30

members of a group or network.

$$y_i = c + \underbrace{\alpha_g}_{\text{correlated effect}} + \beta x_{i,g} + \underbrace{\delta \bar{y}_{-i,g}}_{\text{endogenous effect}} + \underbrace{\xi \bar{x}_{-i,g}}_{\text{exogenous effect}} + \varepsilon_i$$

The model decomposes spillovers to a direct behavioural, or *endogenous* effect, $y_j \rightarrow y_i$, where the outcome of interest among peers affects that of the target individual. Robust applications of the Manski model demonstrate credible approaches to measuring the *endogenous effect*. Damm and Dustmann (2014) analyses movers to new neighbourhoods among individuals through the ‘Danish Dispersal Policy’ of refugee arrivals (1986-1998). Chetty et al. (2016) measures a similar effect among high-poverty children through the US 1990s ‘Moving to Opportunity’ scheme. These studies exploit quasi-randomness in the location assignment of individuals and peers for robust identification, a strength absent from this analysis’ time-invariant dataset only partially mitigated by the use of fixed effects and assumptions of locaiton stickiness.

Additionally, the Manski (1993) model requires contextual or *exogenous* effects, of which the variation of non-outcome of interest variables among peers also directly affects that of the target individual, $x_j \rightarrow y_i$. This brings sophistication to the nature of peer spillovers. A simplified model only allowing endogenous effects through the variable of interest, $y_j \rightarrow y_i$ is nested within this structural model, but not identified due to the ‘Reflection Problem’ and as described in this analysis in section 4.1. In this analysis’ firm productivity model, the *exogenous effect* allows for an additional level of input factor capital and labour to induce higher target TFP, better management practices or more efficient combination of factor inputs, perhaps from observing the mere high level of these factors in their peers, without the peers themselves necessarily experiencing a productivity increase. It is additionally a common specification requirement for every individual to have an own-effect of exogenous characteristics, $x_i \rightarrow y_i$, which corresponds to the standard production function model in this analysis of $k_{i,t}, l_{i,t} \rightarrow y_{i,t}$.

$$\begin{aligned} y_{i,g} &= c + \alpha_g + \beta x_{i,g} + \delta \bar{y}_{-i,g} + \xi \bar{x}_{-i,g} + \varepsilon_i \\ \bar{y}_g &= c + \alpha_g + \beta \bar{x}_g + \delta \bar{y}_g + \xi \bar{x}_g + \bar{\varepsilon}_g \\ (1 - \delta) \bar{y}_g &= c + \alpha_g + (\beta + \xi) \bar{x}_g + \bar{\varepsilon}_g \\ \bar{y}_g &= \frac{c + \alpha_g}{1 - \delta} + \frac{\beta + \xi}{1 - \delta} \bar{x}_g + \frac{\bar{\varepsilon}_g}{1 - \delta} = \bar{y}_{-i,g} \quad (\text{large } g \text{ assumption}) \\ y_{i,g} &= \frac{c + \alpha_g}{1 - \delta} + \beta x_i + \frac{\xi + \delta \beta}{1 - \delta} \bar{x}_{-i} + \frac{\delta}{1 - \delta} \bar{\varepsilon}_g + \varepsilon_i \quad (\text{substituting}) \\ y_{i,g} &= \alpha_g + \pi_0 + \pi_1 x_i + \pi_2 \bar{x}_{-i} + v_g \end{aligned}$$

The focus of Manski (1993)’s linear-in-means (LMM) contribution is to highlight the ‘Reflection Problem’, an econometric trap. The derivation give above shows that the initial structural equation cannot be estimated directly with OLS due to the endogeneity of $\bar{y}_{-i,g} : \mathbb{E}[\bar{y}_{-i,g}\varepsilon_i] \neq 0$. Causal variation in $y_{i,g}$ is unidentifiable due to insufficient reduced-form parameters.

A key solution to this is the deployment of instruments, $z_{-i} \propto \bar{y}_{-i} \cap z_{-i} \not\propto y_i$. An application employing credibly exogenous instruments for a peer effects measure in an individual crime setting is given in Dustmann et al. (2024). The equivalent valid instrument in this analysis which is correlated with the peer effect but has no direct effect on the target outcome of interest is an additional spatial lag of *exogenous* regressors in the static model. This means the capital and labour of a firm k who is a ‘peers of peers’: a peer of firm j , but not of firm i , the last of which is the target firm of interest. In this analysis’ dynamic model, deep temporal lags $y_{-i,t-2} \propto y_{-i,t} \cap y_{-i,t-2} \not\propto y_{i,t}$ are also as a valid instrument under an assumption of weak exogeneity of predetermined regressors.

A modern paper, Bramoullé et al. (2009), explores different identification techniques and their rules for validity for group and network structures. These structures are more complex than the Manski (1993) model, which by using a variable of means $\bar{y}_{-i,g}$, implicitly assumes simple, ‘leave-one-in’ group averages as the peer *endogenous* regressor.

The paper subsequently explores many complex interaction structures summarised by the weighting matrix $\mathbf{W}$, the n -dimensional interactions matrix between firms. I discuss and illustrate structures of $\mathbf{W}$ in detail in section 3.4. The structures of $\mathbf{W}$ are key to identification. The contributions from this paper are, in short, that complex $\mathbf{W}$ matrices which vary group sizes (s_g) and network diameters (s_d) are key to discovering valid spatial lag instruments. Fortunately, these structures resemble realistic interactions common to empirical data, and are applicable to this analysis’ pattern and layout of firm interactions. This strategy continues to assume exogenous assignment of spatial assignments and so target-peer interactions.

The identification rules from Bramoullé et al. (2009) are summarised in Table I. Critically, once certain necessary conditions for the group and network structures are met, an additional spatial lag on the *exogenous* regressors, $\mathbf{W}^2x$, tends to be a valid instrument. This analysis uses Table I’s approaches to instruments to strip out the endogeneity of a spatial peer effects model.

TABLE I
Spatial econometric identification rules for weighting matrix $\mathbf{W}$

Structure	Required condition	Identified	Instrument
<i>Group structure</i>			
Proposition 2	$\xi_0 - \delta_0\beta_1 \neq 0$ and $\zeta_0 - \delta_0\beta_2 \neq 0$		
Linear-in-Means ^a	Target is own peer	×	—
Leave-one-out ^b	Target excluded from peer mean	×	—
	Same group size		
Varying group sizes ^c	Target excluded from peer mean ≥ 2 group sizes	✓	$\mathbf{W}^2x$
<i>Network structure</i>			
Proposition 1	$\mathbb{I}, \mathbf{W}, \mathbf{W}^2$ linearly independent $\beta\theta_0 + \xi_0 \neq 0$	✓	$\mathbf{W}^2x$
Intransitive triad	$\exists i, j, k : \mathbf{W}_{ij} \neq 0, \mathbf{W}_{jk} \neq 0, \mathbf{W}_{ik} = 0$	✓	$\mathbf{W}^2x$
Deep network	$s_d \geq 2$ network diameter	✓	$\mathbf{W}^2x$
<i>Adding correlated effects</i>			
Proposition 4	$\mathbb{I}, \mathbf{W}, \mathbf{W}^2, \mathbf{W}^3$ linearly independent $\xi_0 - \delta_0\beta_1 \neq 0$ and $\zeta_0 - \delta_0\beta_2 \neq 0$	✓	$(\mathbb{I} - \mathbf{W})\mathbf{W}^2x$
Group structure	$s_g \geq 3$ group sizes	✓	$(\mathbb{I} - \mathbf{V})\mathbf{W}^2x$
Network structure	$s_d \geq 3$ network diameter	✓	$(\mathbb{I} - \mathbf{V})\mathbf{W}^2x$

Adapted from Bramoullé et al., 2009 with notation revised and abridgements to improve relevance to this analysis. (a) Manski, 1993, in this model $\mathbf{W}^2 = \mathbf{W}$ for linear dependence. (b) Moffitt and Caspi, 2001, in this model, $\mathbf{W}^2 = \frac{1}{s-1}\mathbb{I} + \frac{s-2}{s-1}\mathbf{W}$ for linear dependence. (c) Lee, 2007. Structures under a **Proposition** header must follow the **Proposition**'s required conditions. x denotes an *exogenous* regressor, in this firm analysis this means $k_{i,t}$ or $l_{i,t}$. It is comparable to the vector of regressors $\mathbf{x}_{i,t}$ that I use to denote the vector of both of them.

3. DATA ANALYSIS

3.1. FAME

I use data from Fame for this investigation. Fame is a comprehensive database of UK firms with balance sheet variables, which can be summed to useful aggregates for economic analysis: GVA, capital, and labour inputs. It is provided by Moody’s Analytics, an international credit agency, and collects its data from UK firms via survey¹¹. Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, given the background discussion laid out, the UK dataset is sufficiently comprehensive and policy-relevant.

Over 100 GB of raw data were accessed and downloaded from Fame in February 2025 by Professor Lars Nesheim for Irastorza-Fadrique et al. (2026) and reused for this project. Raw data were ingested to *DuckDB*, an open-source, MIT-licensed relational (SQL) database architecture¹² optimised for high-compute variable manipulation. This is crucial for the applications of $(n \times n)$ matrix transformations to a variable n -length column vector, as I detail heavily for calculating peer weights and high-order transformations.

Every firm is tagged with raw, textual location properties. I construct a hierarchy of favourability to parse these, in order: *latitude/longitude*, *full address*, *address* (split over multiple fields), and *postcode* only. I additionally favour data listed for the firm’s *primary trading address* over its *registered office address* (headquarters). When only postcode data was available, I matched postcodes to the ONS’ postcode lookup database¹³, yielding a decimal longitude-latitude value for the centre of its area. I compiled these heterogeneous features to 7 location identifier columns. The first set of these indicate membership of 3 spatial location groups: travel-to-work area (*ttwa*) using the 2011 definitions¹⁴, the first-half of a UK postcode (*pc4*), or a full UK postcode (*pc8*). The latter 2 are point distance decimals for longitude + latitude. *longitude_decimal*, *latitude_decimal*, and 2 supporting columns indicating the data quality.

The resulting time-invariant table has exactly 152,379 firms with unique Companies House *registered_number* identifiers.

¹¹ Moody’s Analytics (2025)

¹² Mark Raasveldt and Hannes Muehleisen (2026)

¹³ Office for National Statistics (2025)

¹⁴ Travel-to-work area definitions from 2011 are still considered the mostly reliable for current analytical research, due to the pollution in travel and working patterns resulting from the COVID-19 pandemic during the 2021 Census. Travel-to-work-area area coverages are mostly time-invariant; changes between 2001 and 2011 were reasonably limited, see: Office for National Statistics (2015).

3.2. DERIVING PANEL AGGREGATES

All financial data are deflated to real terms, using 2024 as the base year. I use CPIH¹⁵ to deflate *renumeration_employees* and other income-related aggregates capturing household spending patterns. I use the GDP deflator¹⁶ for all other balance sheet variables. Financial data are in thousands of GBP (£'000s); as I estimate log relationships exclusively, this has no effect on regression coefficients. I compute these variables for the exactly 1,128,490 panel observations. Firms have a mean of 7 years of observations per firm. Further detail on the cleaning procedure and derivation of variables is available in Appendix A.1.

Labour input is the number of employees reported on the FAME balance sheet per firm. This dataset does not include firm-level hours worked per employee that would offer granular productivity (GVA per hour worked) measures. An estimation in this setting would require imputing the measure from industry aggregates, which would likely confound the true firm-level signal.

Capital is Fame's *fixed_total* variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm's fixed capital. The average capital-output ratio is 6x within FAME when filtering for positive values Table II, which is moderately higher than macroeconomic aggregates typically suggest likely due to accounting standard differences.

TABLE II
Capital-output ratios in FAME corpus of firms

Variable	Description	Capital-output ratio
<i>tangibles</i>		3.53
<i>fixed_total</i>	tangibles + intangibles + inv._other	6.00
<i>total_assets</i>	fixed_total + current_assets	11.56

GVA is calculated as the sum of *ebitda* and *renumeration_employees*. *renumeration_employees* is the exact sum of *wages*, *social_security_costs*, and *pensions_costs* in the balance sheet. *ebitda* is in turn *turnover* minus intermediate inputs and is given in the Fame balance sheet for the vast majority of panel observations in our dataset. It is also possible to reconstruct *ebitda* via other balance sheet metrics, but these sub-fields are actually available for fewer observations than *ebitda*.

¹⁵ Office for National Statistics (2026a), CPIH is the ONS' preferred price index for household consumption Office for National Statistics (2026b)

¹⁶ Office for National Statistics (2026c)

To help guide the choice of aggregate measures and understand the observation structure, I conduct a benchmark 2-factor production estimation in panel structure to help validate the choice of aggregates against known measurements.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + u_{i,t} \quad (1)$$

This specification does not yield consistent estimates when estimated with OLS, according to the structural model I give in section 4.1. $u_{i,t}$ is a non-structural error-residual correlated with $k_{i,t}$ and $l_{i,t}$. Nevertheless, it is a helpful exploration tool for parameter selection.

TABLE III
Benchmark 2-factor TFP estimation

	(1.) <i>gva</i>	(2.) <i>gva</i>	(3.) <i>gva</i>	(4.) <i>gva</i>	(5.) <i>gva</i>	(6.) <i>gva</i>
constant	2.995 (0.036)	3.036 (0.031)	1.970 (0.010)	1.973 (0.010)	4.751 (0.018)	5.268 (0.070)
<i>employees</i>	0.617 (0.005)	0.614 (0.004)	0.561 (0.002)	0.561 (0.002)	0.720 (0.004)	0.628 (0.015)
<i>total_assets</i>	0.286 (0.005)	0.284 (0.004)	0.421 (0.002)	0.421 (0.002)		
<i>fixed_total</i>					0.065 (0.002)	
<i>tangibles</i>						0.090 (0.006)
<i>intangibles</i>						0.025 (0.002)
<i>investments_other</i>						0.007 (0.001)
<i>i</i> fixed effects	✓	✓			✓	✓
<i>t</i> fixed effects	✓		✓		✓	✓
Observations	1,045,164	1,045,164	1,045,164	1,045,164	1,006,904	152,444
R ² (excl. F.E.)	0.3313	0.3453	0.8094	0.8096	0.2830	0.2699
R ² (Overall)	0.7912	0.7900	0.8096	0.8096	0.6751	0.7403

Coefficients derived in Table III give a labour income share, $\hat{\beta}_2$, exactly in line with ONS analysis of UK industry-level data over the same time period, particularly *corporate gross labour share* which is constructed from a similar sample to Fame¹⁷.

I explore specifications which omit fixed effects. In Model (2.), the capital share of income jumps on the omission of firm fixed effects as all within-firm variation that we would decompose as $\alpha_i \in \varepsilon_{i,t}$ is instead attributed to capital. This gives us some insight that in this panel setting, the measure of capital as a stock variable is reasonably consistent within a firm over time.

¹⁷ Office for National Statistics (2024)

When varying regressor measures, surprisingly, running the regression with *fixed_total* as the capital measure instead of *total_assets* almost entirely eliminates the capital share of income, or explanatory power of capital on variation in GVA. Similar results are shown for a further decomposition of *fixed_total* to its asset components. I leave it for future analysis why this dataset yields this phenomenon and for the rest of the analysis, I use *total_assets* as the primary capital measure.

Helpfully, estimates of the factor income shares for contemporaneous labour and capital stay roughly constant throughout all specifications of this analysis even as other derived parameters significantly jump around. This suggests the effect being measured by the spatial spillover is truly orthogonal and distinct from the classical production model core to the usual theory of the firm.

3.3. FINANCIAL AND LOCATION DATA LIMITATIONS

Following data cleaning procedures discussed in Irastorza-Fadrique et al. (2026), I removed duplicate entries, limited observations to those with more than 10 employees, and observations with the required balance sheet properties, *renumeration_employees* and *ebitda* required to calculate firm GVA.

Fame has rich microdata observations with granular detail about each firm in the UK, but it is not designed for spatial analysis without advanced data science techniques, improving even beyond the sophistication demonstrated here. There is some miscoding in Fame’s location data. Despite it being my favoured *amalgamated address* selection choice, Fame’s native *longitude* and *latitude* data contained some errors, such as recording either one of those 0.000 coordinates, swapping them, or placing them outside the UK. I employed several techniques to search for outliers. One instances is calculating the centre-point of all firms sharing a postcode and then filtering for firms further than a threshold: I evaluated that 1.8% of all firms in the cleaned sample are more than 5km away from their postcode’s centre-point. Due to the time and data processing constraints of this analysis, I dropped these firms and observations.

Though I recorded the quality of location data, I undertake no further analysis using it as its incorporation. I discuss further uses in section B.1 to control and investigate the effects of miscoded location data and heterogenous treatment effects. These requires complex data science and Machine Learning approaches, which I leave for future analysis.

I amalgamate multisite firms to their primary trading address, as an assumption. In the dataset prepared for Irastorza-Fadrique et al. (2026), each firm is only tagged with their last-known registered address and primary trading address. Fame includes further trading addresses for multisite firms available for download through their data portal, however the only site-level datapoint provided is *num-*

ber_of_employees per trading address, with sparse and inconsistent coverage¹⁸. This analysis simply takes financial aggregates listed for each firm and assumed it to be located at the listed primary trading address. There are clear flaws to this method: large retail firms, such as consumer retailer ‘Tesco’, have several hundred listed trading addresses on Fame, and generate much less than the full amount of GVA at their listed primary trading location in ‘Welwyn Garden City’. Nevertheless, difficult algorithm allocation decisions, or the pursuit of more sophisticated data sources are required to address this issue, as discussed in section B.4. The amalgamation assumption provides ease of computation while contributing to attenuation bias of the estimated spillover parameters due to imperfect peer assignment.

Additionally, the location data that is available is time-invariant, a characteristic of the *fixed* table, rather than updated on a yearly basis to reflect firm-movers. This is a reflection of using a comprehensive dataset of financial datapoints, that was not optimised for spatial analysis; instead Fame provides an in-built feature suggesting peers via balance sheet and industry similarity, which I ignore for this analysis. One interpretation of this analysis’ use of these characteristics is that each firm is over the entire 20-year sample with no movers. This seems very unlikely: ?? ‘??’ tells us to expect the endogenous location of firm movement and colocation to optimise for taking advantage of firm spillovers.

Using the location characteristics as I do, as a noisy signal of the firm’s true location, has 2 econometric consequences. Firstly, firm fixed effects in all spatial specifications become extremely important. For all movers within the sample period, an endogenous firm location decision can only be modelled as correlated with the spillover regressors of interest. This justifies my use of fixed effects over random effects throughout this analysis. Firm fixed effects are time invariant, and can additionally help to eliminate some of the endogeneity induced by movers who undertook their move just before the sample period, 2006-, to maximise their spillover effect.

Secondly, I acknowledge some attenuation to the productivity spillover estimate. Incorrect location data due to any of the above-discussed issues causes the assignment of spatial peers to any target to be miscalculated, and vice versa. Peer firms who were close to the target firm during the firm-year observation who subsequently moved away within-sample will have an extreme distance recorded, and so be excluded from tagged address groups and truncated distance calculations. Peer firms who are recorded as close to the target firm as they subsequently moved to the location. This reduces the magnitude of the spillover coefficients in θ_0 estimated.

¹⁸ Moody’s Analytics (2025), platform accessed August 2026.

3.4. LOCATION GROUP CLASSIFICATION

Throughout this analysis, I use terminology of a productivity spillover occurring from a group or average of *peer* firms onto a *target* firm. The influence of the vector of all firms j on the vector of all firms i can be summarised in an $(n \times n)$ interactions matrix, $\mathbf{W}$. A weight $w_{ij} \in \mathbf{W} \neq 0$ implies a firm-to-firm interaction, namely some use of the peer firm's attributes in the regression to estimate the target firm's TFP.

This interactions matrix is symmetric: any firm with a peer weight calculated by the distance between the two firms naturally is a peer in return with an identical distance and weight. Extensions to this analysis could devise asymmetric relations, for example, with bigger firms having more influence. The interactions relationship is irreflexive - no target can be a peer of itself. This means the diagonal elements of $w_{ii} \in \mathbf{W} = 0$. I summarise the properties for $\mathbf{W}$ in Table IV, with further discussion for $\mathbf{W}_g$ and $\mathbf{W}_d$ to follow.

TABLE IV
Weighting matrices i, j -relation properties

Matrix type	Symmetric	Reflexive	Transitive	Overlapping	Blocks
Group, $\mathbf{W}_g$	✓	X	✓	X	g
Network, $\mathbf{W}_d$	✓	X	X	✓	$ttwa$

The definition of a group within $\mathbf{W}_g$ is intuitive: peers interact in a fixed limit of nearby firms, where each peer within the group is treated as homogenous, offering an equal weighted value, here $1/(s_g - 1)$. The group matrix $\mathbf{W}_g$ is transitive: if firms A and B share a spatial group, $ttwa$, $pc4$, or $pc8$, then a firm C that is a peer of firm B must also be of firm A. Apart from its irreflexivity, $\mathbf{W}_g$ is block diagonal. I construct initial weights by peers which share the spatial group $pc8$, so each firm has peers of all the firms in its postcode, each weighted with the reciprocal of the number of firms in the postcode (leaving-one-out).

Formal rules for construction of the group matrix are summarised below¹⁹. I illustrate an example $\mathbf{W}_{g*}$, a (7×7) set of interactions between firms which have group classifiers of size $s_1 = 4$ and $s_2 = 3$ only in Figure I.

¹⁹ Adapted from Bramoullé et al. (2009)

$$w_{ij} \in \mathbf{W}_{g^*} = \begin{cases} 0, & \text{if } i = j \\ \frac{1}{s_g - 1}, & \text{else if } (i, j) \in g \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbf{\Gamma}_{s_1} \in \mathbf{W}_{g^*} = \{i, j \in g\}, s_1 : n_g$$

$$\mathbf{W}_{g^*} = \begin{pmatrix} \mathbf{\Gamma}_{s_1} & 0 & \dots \\ 0 & \mathbf{\Gamma}_{s_2} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}_{(n \times n)}$$

FIGURE I
Illustration of group matrix $\mathbf{W}_g$

$$\mathbf{W}_{g^*} = \begin{pmatrix} 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$$

Weighting matrix $\mathbf{W}$ illustration for a network of 7 interacting firms in a group *pc8*. Of these, firms 1-4 share their postcode, as do firms 5-7.

I can extend this to more complicated set of group rules. I calculate non-overlapping group weights $\mathbf{W}_g$ to $\mathbf{W}_{g1}, \mathbf{W}_{g2}, \mathbf{W}_{g3}$ at once, which represent the successive concentric rings of *pc8* (core), *pc4*, and *ttwa*. The weighting rules for these multiple, non-overlapping groups is to zero the i, j interaction in W_{gk} if a prior matrix in the series, W_{gl} , $l < k$ has a non-zero entry in that cell. I illustrate an example in Figure II. These matrices are orthogonal.

$$w_{ij} \in \mathbf{W}_{gk} = \begin{cases} 0, & \text{if } i = j \\ 0, & \text{if } w_{ij,l} \neq 0, \exists l < k \\ \frac{1}{s_g - 1}, & \text{else if } (i, j) \in g \\ 0, & \text{otherwise} \end{cases}$$

Firms have been labelled using their location features to symmetric, transitive, non-overlapping groups by *ttwa* (largest) to *lat_lon5* (smallest), with the latter meaning two firm entries shared a latitude-longitude identifier to 5 decimal places. Reviewing the summary statistics, I opted for full postcodes, *pc8* as the group

TABLE V
Summary statistics across location groups

Property	#groups	Unclassified	Unclassified (%)	s_g avg. peers per group
<i>ttwa</i>	228	567	0.4%	668.3
<i>pc4</i>	2,705	0	0.0%	56.3
<i>pc8</i>	49,963	0	0.0%	3.1
<i>lat_lon3</i>	48,540	507	0.3%	3.1
<i>lat_lon4</i>	53,193	507	0.3%	2.9
<i>lat_lon5</i>	53,246	507	0.3%	2.9

FIGURE II
Illustration of multiple, non-overlapping groups

$$\mathbf{W}_{g1:3^*} \sim \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & 0 & 0 & 0 \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & 0 & 0 & 0 \end{bmatrix}$$

Non-overlapping group illustration for a network of 8 interacting firms in a *ttwa*. Of these, firms 1-5 share the first half of a postcode - are in a *pc4* group, as are firms 6-8. Within those, firms 1-3 share a full postcode, as do firms 7-8. Firms 4-6 are isolated in their *pc8* groups.

identifier for initial 1-group models in Table VI. This was mostly driven by full data availability, and the consistency of postcode definitions with the larger group *pc4*. Deriving a subgroup by latitude-longitude match within the postcode seemed fruitless, as in many cases it would identify the exact same firms, a move from the average number of peers from 3.1 to 2.9, and was a similar number of groups, with non-cohesive definitions.

3.5. LOCATION DISTANCE CALCULATIONS

A network matrix is a more complex set of interactions where each individual i has their own set of peers by whom they are influenced. In the general definition (Bramoullé et al., 2009), the influence of each individual’s peer set is still the simple ‘leave-one-out’ average of their heterogenous peers, however, differing weights applied within the peer set are equally permissible. I use variants on the reciprocal of the spatial distance between the two firms to construct my weights in $\mathbf{W}_d$ in this analysis. Intuitively, these distance reciprocal weights are pretty similar. I explore 3 models just to demonstrate that findings are robust to small modifications in the weights, and that differing weighting strategies are permissible. A simple reciprocal weight is common to the gravity models used in the urban economics literature discussed²⁰. A harsher penalty for distance, $\frac{1}{\delta^2}$, has been employed by the New Economic Geography literature, emphasising a more crucial role for close proximity²¹.

$$w_{ijt} \in \Omega_t = \begin{cases} \frac{1}{d_{ij}}, & (1) \text{ gravity} \\ \frac{1}{d_{ij}^2} & (2) \text{ N.E.G.} \\ \exp -d_{ij} \cdot k & (3) \text{ exponential decline} \end{cases} \quad (2)$$

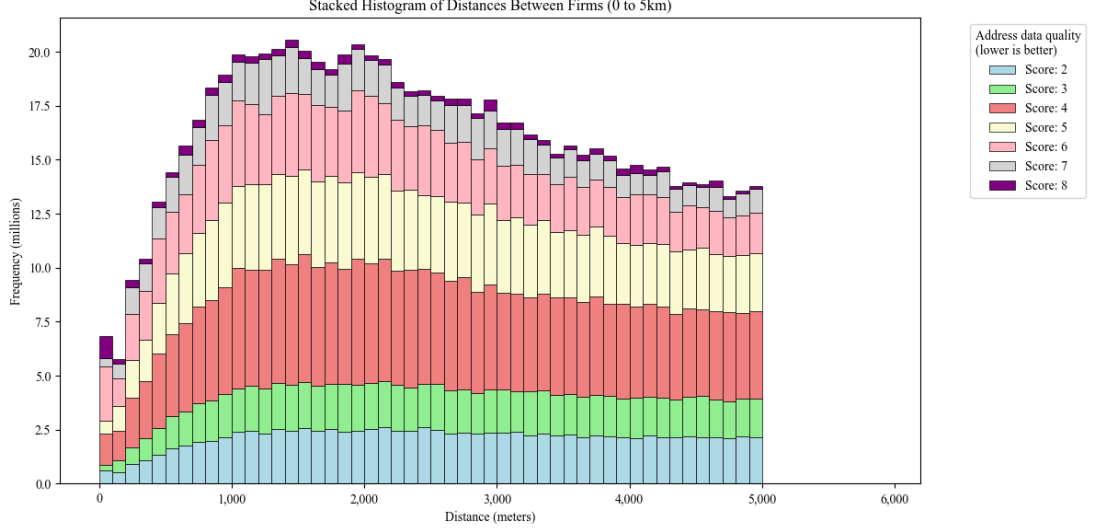
Distances are calculated as the Haversine distance between any 2 firms. This represents the straight-line, ‘as-the-crow-flies’, fastest path between the longitude and latitude pairs given, which has a slight adjustment from a standard straight-line distance due to the (great-circle) curvature of the Earth, for which the effects are minimal due to the small distances calculated. I show a histogram of distances calculated between firms in Figure ??.

To reduce computational complexity, I disregard any possible matches which don’t share a Travel-to-Work-Area (*ttwa*). Not only are these firms more likely to be further apart, eliminating the need to calculate distances between, say, London and Glasgow, the *ttwa* boundaries also a useful proxy for the functional boundaries of interacting economic agents, so it is a reasonable restriction to

²⁰ (Gravity)

²¹ (NEG)

FIGURE III
Histogram of distances between any target-firm pairing



Plotted 0.1% sample of one-to-one relational table of distance calculations, into bins of 100m. Using the $d1$ distance measure, which is hard-bounded that no firm can have a firm-peer spillover relation past a $ttwa$ boundary, truncated at 5m. Colours indicate differing levels of data quality for addresses within Fame. Frequency in millions of target-firm connections.

limit the peer effects of firms across those borders. Additionally, I disregard any spatial connection above a 5km limit. Practically, this saves on memory and storage. It aligns helpfully with the limit to when contributions to a weighted average aggregate start to become infinitesimal using a reciprocal function for weights. Conceptually, the threshold provides a plausible limit for the practical boundaries of one firm's peer effects on another and is the lower limit of the literature consensus²². This is a crucial additional assumption, as discussed further at the end of this section. Figure III shows that the 5km boundary is a noticeable truncation of all firm-to-firm connections within a $ttwa$.

One final helpful property within the $\mathbf{W}_d$ matrix is the definition of a network itself. A network is the total set of individuals who have at least one interaction within the network:

$$j \in N_d \iff \mathbf{W}_{ij} \neq 0, \exists i \in N_d$$

There can exist several networks within a $\mathbf{W}$ interactions matrix, as certain individuals within a peer set may have these complex interactions, but all inside the set have nothing with those outside. Organising these networks in a $\mathbf{W}_d$ gives an outer block-diagonal structure, though with far larger blocks than $\mathbf{W}_g$. By disregarding all firm pairs which don't share a $ttwa$, I set the limits of a network

²² Graham and Gibbons (2018), p.22

to the *ttwa* level. Networks offer further calculation opportunities for network fixed effects as discussed in section B.3.

$\mathbf{W}_d$ is non-transitive. Firm B, a peer of firm A within its 5km radius, can clearly have its own peers that are within its own 5km radius that are beyond that radius for firm A. This matrix resembles a sparse matrix with a ‘thick’ diagonal, rather than a matrix with distinct large blocks. I illustrate a network matrix using real sampled data from the cleaned dataset in Figure IV.

FIGURE IV
Illustration of distance-weighted network matrix $\mathbf{W}_d$

$$\mathbf{W}_{d^*} = \begin{bmatrix} 0 & 0.42 & 0.29 & 0.28 & 0 & 0 & 0 & 0 & 0 \\ 0.31 & 0 & 0.38 & 0.32 & 0 & 0 & 0 & 0 & 0 \\ 0.09 & 0.17 & 0 & 0.74 & 0 & 0 & 0 & 0 & 0 \\ 0.09 & 0.14 & 0.76 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.36 & 0.27 & 0.36 & 0.27 \\ 0 & 0 & 0 & 0 & 0.02 & 0 & 0.02 & 0.96 & 0.02 \\ 0 & 0 & 0 & 0 & 0.24 & 0.38 & 0 & 0.38 & 0 \\ 0 & 0 & 0 & 0 & 0.02 & 0.96 & 0.02 & 0 & 0.02 \\ 0 & 0 & 0 & 0 & 0.24 & 0.38 & 0 & 0.38 & 0 \end{bmatrix}$$

Illustrative 9×9 $\mathbf{W}$ network structure generated by taking a sample from the cleaned Fame data of a random block of 4 firms sharing a *ttwa*, and a random block of 5 firms. True $\mathbf{W}_d$ matrices have *ttwa* have much larger blocks within the block diagonal structure. Weights use the *d1* algorithm, which is the reciprocal of *distance_meters* row-normalised. Values rounded to 2 decimal places.

To calculate $\mathbf{W}_d$, I store distances d_{ij} in a one-to-one relational table, and compile rows on an efficient as-needed for calculations. The distance table has 734 million target-peer pairs; paradoxically, this implies both an incredibly rich, high-identification, interactions table due to a mean of 4,820 peers per target firm and also that the $\mathbf{W}_d$ matrix is mostly (over 96%) sparse.

The *ttwa* and 5km boundaries are ultimately an arbitrary choice; stricter restrictions to create networks only within *pc8* boundaries would have resulted in a $\mathbf{W}_d$ matrix that is 99.8% sparse. While these could be considered reasonable restrictions, they matter a lot to the econometrics. Firms disqualified as peers are not merely excluded from the weighted average calculations, but are considered valid candidates for instruments with exclusion restriction, as we assume they are ‘too distant’ to have a direct effect on the productivity of the target firm i ; they are merely non-transitive peers-of-peers. This is therefore a crucial boundary: errors in the specification of this parameter could lead to Incorrect specification of the peer group of the target firm. I discuss in section B.1 how the most appropriate approach would be to cross-validate these non-linear parameter selections in a choice regression specification, minimising mean-squared-error (or adjusted R^2).

3.6. HIGHER-ORDER $\mathbf{W}$ TRANSFORMATIONS

Intuitively, I take the product of these weighting matrices $\mathbf{W}$ to the aggregate variables vectors *gva*, capital *total_assets*, and labour *employees*. As will be shown in section 4 ‘Static models’, the procedure to generate valid instrumental variables also requires me to apply multiples of this transformation matrix, exponents $\mathbf{W}^2, \mathbf{W}^3, \mathbf{W}^4$, and as an extension, the higher-order cross-transformation for group matrices, $\mathbf{W}_{g1}\mathbf{W}_{g2}, \mathbf{W}_{g1}\mathbf{W}_{g3}$ and so on to the aggregates.

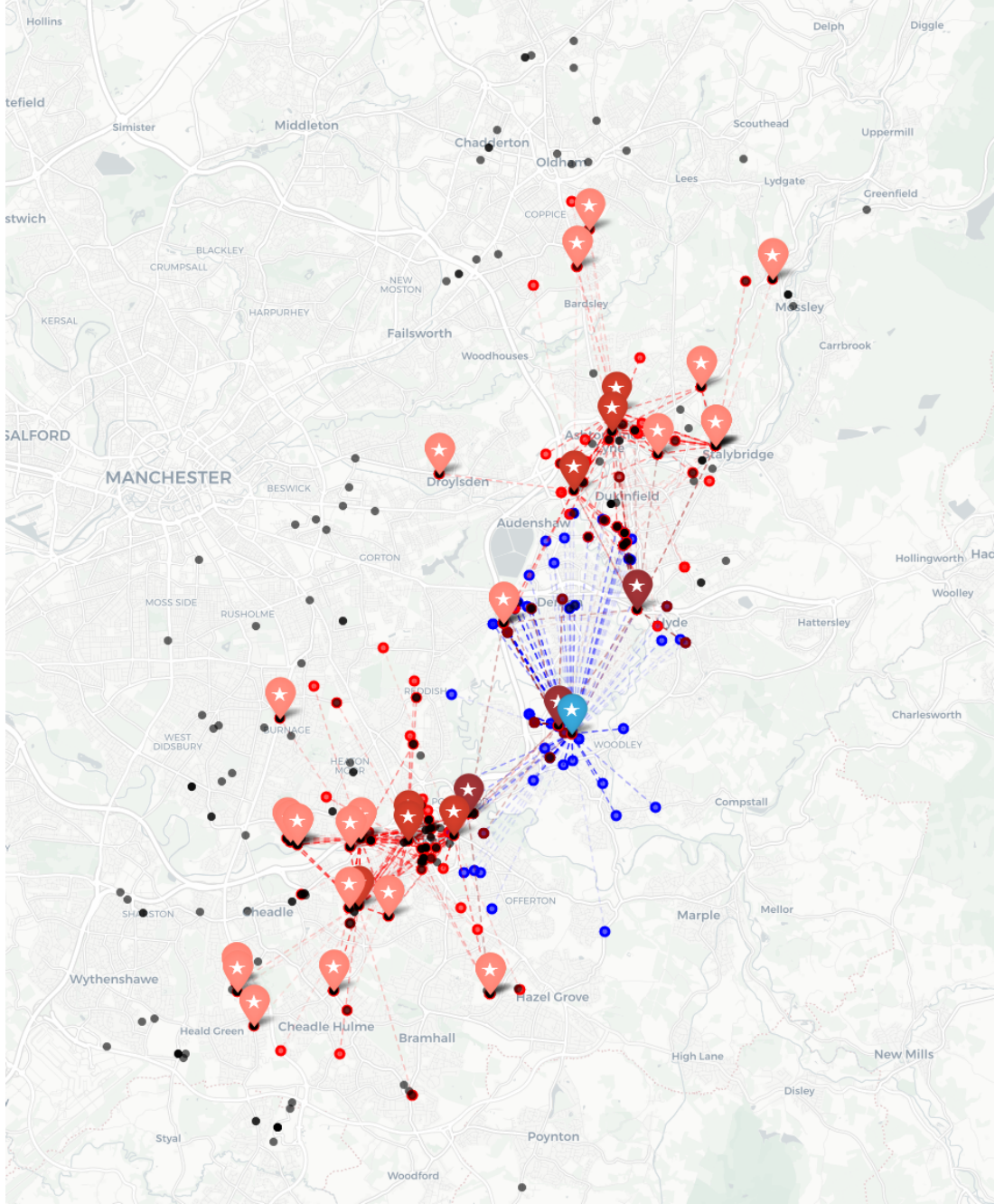
I calculate these for my 3 aggregate variable vectors of interest. Summary statistics offering a sense of the shape of transformed data are available in Appendix B.5.

As reviewed in 2.2, the key condition for discovery of valid instruments in this peer effects context is the linear independence of $\mathbb{I}, \mathbf{W}, \mathbf{W}^2$. With the additional step of application of network fixed effects (section 47), this is additionally evaluated with $\mathbf{W}^3$. I can trivially say this is true, meeting the alternative sufficient conditions in I of ≥ 3 group sizes for the group $\mathbf{W}$ matrices.

My distance-decay weights, $\mathbf{W}_{d1}, \mathbf{W}_{d2}, \mathbf{W}_{d3}$ meet the strong alternative $s_d \geq 3$ condition, where s_d is the *diameter* of the network, the maximal number of links connecting any target i with another firm through a series of peers $j, k, \dots$. In the analysis of this set of firms with distance radius limits, the *diameter* is large, potentially spanning the full spatial diameter of a *ttwa*. It is this mechanism design provides another justification, alongside computation, for limiting the max length of a target-firm peer relation to 5km: it vastly increases the probability of ‘overlapping groups’, i.e. a high diameter network within a *ttwa*. The overlapping group network structure lends itself nicely to illustrations. Figure V selects a random firm i from my sample, which happens to be in Stockport, Greater Manchester, and draws the distances used to calculate metric $d1$ between this target and its peers. In typical network diagram fashion, a network with a considerable *diameter* is created by taking a subset of i ’s peers, then successively a set of their peers, then successively some of those and so on. 4 iterations in this diagram fails to clearly hit the *ittwa* boundary, on the south side, with more argument for this behaviour on the north side where peer nodes randomly selected may be starting to bunch at an ‘invisible’ border of the *ttwa* boundary.

FIGURE V

Illustration of overlapping groups from a target firm to its whole network through successive peers



(blue pin): target firm i with a full set of its (blue dot) peers j , distances d_1 metric drawn in.
 (dark red pins): sample (limit 5) of j peers, then calculated as their own targets
 (red pins): sample (limit 5) of k peers of firms j , some of which are also a peer of i .
 (light red pins): sample (limit 5) of m peers of firms k , few of which are also a peer of i .
 (light red dots): set of n peers of firms m , few of these are peers of i, j , probabilistically creating a network diameter of $s_d \geq 3$. Plotting software from Folium (2012). Map tiles from CARTO (2026).

4. STATIC MODELS

I use a stylised 3-factor production model throughout this analysis. Y , A , K , and L are each firm-year observation's GVA, TFP, labour, and capital in levels. I denote logged terms with lower-case notation and vectors in bold.

Log TFP, denoted by $z_{i,t}$, is the residual after the own-firm contribution of the two classical inputs. I decompose it to a constant across all observations, firm and time fixed effects and, most crucially for this investigation, a vector of the weighted sum of all peer 3-factor production factors with the dot product against the vector of their respective spillover coefficients $\boldsymbol{\theta}_0$:

$$\mathbf{z}_t^{spill} = \left(\sum_{j \neq i} w_{ij} z_{j,t} \quad \sum_{j \neq i} w_{ij} k_{j,t} \quad \sum_{j \neq i} w_{ij} l_{j,t} \right) \quad (3)$$

$$\boldsymbol{\theta}_0 = \begin{pmatrix} \delta_0 & \xi_0 & \zeta_0 \end{pmatrix}' \quad (4)$$

Different static weights, w_{ij} , are explored throughout this section. The objective is to estimate the vector of structural coefficients $\boldsymbol{\theta}_0$ of spillover from peers to any given target firm.

$$Y_{i,t} = A_{i,t} \cdot K_{i,t}^{\beta_1} \cdot L_{i,t}^{\beta_2} \quad (5)$$

$$z_{i,t} = \ln(A_{i,t}) = y_{i,t} - \beta_1 k_{i,t} - \beta_2 l_{i,t} \quad (6)$$

$$= \alpha_i + \gamma_t + \beta_0 + \mathbf{z}_t^{spill} \boldsymbol{\theta}_0 + \varepsilon_{i,t} \quad (7)$$

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} \quad (8)$$

$$+ \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \xi_0 \sum_{j \neq i} w_{ij} k_{j,t} + \zeta_0 \sum_{j \neq i} w_{ij} l_{j,t} + \varepsilon_{i,t}$$

$$= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t}$$

$$+ \delta_0 \sum_{j \neq i} w_{ij} \cdot (y_{j,t} - \beta_1 k_{j,t} - \beta_2 l_{j,t}) + \xi_0 \sum_{j \neq i} w_{ij} k_{j,t} + \zeta_0 \cdot \sum_{j \neq i} w_{ij} l_{j,t}$$

$$= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t} \quad (9)$$

$$+ \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_0 - \delta_0 \beta_1) \cdot \sum_{j \neq i} w_{ij} k_{j,t} + (\zeta_0 - \delta_0 \beta_2) \cdot \sum_{j \neq i} w_{ij} l_{j,t}$$

Equation (9) is a linear peer effects specification as given by Manski (1993). It is helpful to borrow the terminology from this discussed in section 2.2 and keep in mind the conceptual distinction between these and the more general and related concepts of endogeneity, exogeneity, and correlation. Variables $(k_{i,t} \quad l_{i,t}) \in x_{i,t}$ can be thought of as *exogenous* characteristics of firm i causally affecting its outcome $y_{i,t}$. There is an *endogenous* effect directly from peer outcomes $y_{j,t}$ with coefficient δ_0 . Peers also have *exogenous* characteristics $x_{i,t}$, which have a direct

exogenous effect on the target's productivity $x_{i,t}$ via ξ_0 and ζ_0 . For this section of models, I assume there are no *correlated* effects and leave further discussion of the implementation of these to subsection B.3 'Network fixed effects'. The use of panel fixed effects has no effect on econometrics discussed in section 2.2.

It is important to note that I do not impose restrictions on θ_0 . I allow each of the 3 factors of production to independently exert a spillover effect in general form. This allows this analysis' investigation as to which of the production factors matters more for productivity spillovers, and in what ratio, conceptually linked to the discussion in ?? about identifying the channels through which productivity spillovers might occur. Through this 3-factor production model, it is hard to link each parameter in θ_0 to a Marshallian channel, but broadly we might consider ζ_0 , the coefficient on an increase in a peer firm's labour inputs affecting a target's TFP to resemble Marshall (1920)'s labour pooling, with some interaction to knowledge spillovers, and δ_0 , the coefficient from a peer firm's TFP to target TFP to be the remainder of knowledge spillover effects. A TFP spillover occurring from peer firm's capital ξ_0 would be a novel channel to Marshall (1920)'s discussion and could reflect the idea of more capital-intensive peers. Regardless, it is strongly justifiable that we would expect the coefficient on each peer production factor to spillover to a target firm's production heterogeneously, and not be proportionate to the contribution of these factors to their own-firm GVA. I have no *a priori* arguments for the sign of the θ_0 coefficients and their relative magnitudes compared to their contributions to own-firm GVA in β_1, β_2 .

$$y_{i,t}^{(y)} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} + \varepsilon_{i,t} \quad (10)$$

$$y_{i,t}^{(z)} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (11)$$

$$= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} - \delta_0 \beta_1 \sum_{j \neq i} w_{ij} k_{j,t} - \delta_0 \beta_2 \sum_{j \neq i} w_{ij} l_{j,t}$$

Two alternative specifications may be proposed: equation (10) modifies (8) to argue that it is the whole of GVA which matters the peer-target spillover, not each component heterogeneously. It is trivial to see that this implies the coefficient on the *exogenous effect* is zero, such that $\xi - \delta_0 \beta_1 = \zeta - \delta_0 \beta_2 = 0$. This has the interpretation alluded to above, that the peer effect of any classical factor, capital and labour, is exactly in the ratio to its own factor income, which is a strong restriction to impose. We might expect this to be merely a nested specification of (9), for which the hypothesis of the zero coefficients can be tested via Wald-test as a joint hypothesis. In actual fact, without strict assumptions on the orthogonality of peer errors, this specification's regressors are fully endogenous, and it falls into

the ‘Reflection Problem’ of identification outlined in section 2.2.

The alternative in equation (11) argues that only the TFP component of production matters for the spillover to influence a target firm’s TFP. This is also a restriction on the more general (8) such that $\xi_0 = \zeta_0 = 0$. At first glance, this may appear a feasible restriction, since it appears this model also follows the structure of the peer effect class of models and is in fact over-identified, with both the coefficients of the peer classical factors and $\sum_{j \neq i} w_{ij} y_{j,t}$ identifying δ_0 . However, the next section 4.1 demonstrates that the reduced form of this approach yields the need for an *exogenous effect* set of restrictions, which coincides with $\xi_0, \zeta_0 \neq 0$. The specification with this TFP-only spillover restriction cannot be identified.

While the classical production function (8) frequently takes the assumption of constant returns to scale, $\beta_1 + \beta_2 \leq 0$, no such restrictions for the firm occur with the endogenous decomposition of TFP. A restriction on logged production (8) collectively on β_1 , β_2 , and θ_0 is difficult to justify. As with other external effects, there is no reason to believe that market factor prices and the firm decision-rule for spatial allocation in view of these spillover effects, $w_{ij} z_{j,t}$, is socially optimal. A peer firm j with extremely high and growing components of GVA has little interest in siting themselves in the centre of a cluster of firms that are struggling with TFP growth, yet the external impacts of that firm would be significant.

4.1. STATIC PANEL IDENTIFICATION

Consistency requires the plausible exogeneity of regressors used in structural equation (9) from the error term $\varepsilon_{i,t}$. Capital and labour inputs in every firm i, j are assumed to be strictly exogenous to every other firm’s TFP error shock, for every period in this static model. This is crucial for consistency with the direct own-firm $k_{i,t}, l_{i,t}$ inputs, but also the use of $\mathbf{w}_{ij} k_t, \mathbf{W}^k l_t$ as an *exogenous* effect and, as we will later see, the ability to leverage higher-order spatial lags of the *exogenous* effect as an unconfounded instrument for $w_{ij} y_{j,t}$. This is a restrictive assumption, but standard in the practice of constructing static panel models. It is relaxed in Section 5.

$$\mathbb{E}[\varepsilon_{i,t} | (k_{i,t} \quad l_{i,t})] = 0_{(1 \times 2)}, \quad \forall i, \forall t, \tau \quad (12)$$

The *endogenous* effect regressor $\sum_{j \neq i} w_{ij} y_{j,t}$ is, unsurprisingly, endogenous, making valid estimation in this setting more complicated. To show the endogeneity requires stacking w_{ij} target-peer weights to an $n \times n$ interactions matrix, $\mathbf{W}$. I

invert the *endogenous* component to give a $(\mathbb{I} - \delta_0 \mathbf{W})^{-1}$ coefficient on all variables.

$$\delta_0 < 1 \implies |\mathbb{I} - \delta_0 \mathbf{W}| \neq 0 \quad (13)$$

The inversion is permissible because I can expect a restriction to apply, ruling out iteratively increasing shocks²³. We can interpret this as a ‘spatial lag polynomial’, directly comparable to the lag polynomial and MA inversion procedure in autoregressive time series analysis²⁴.

$$\mathbf{y}_t = \boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \quad (14)$$

$$+ \delta_0 \mathbf{W} \mathbf{y}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t$$

$$= (\mathbb{I} - \delta_0 \mathbf{W})^{-1} \left(\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \right. \\ \left. + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \right)$$

$$= \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \left(\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \right. \\ \left. + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \right) \quad (15)$$

$$= \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \boldsymbol{\alpha} + \frac{\gamma_t + \beta_0}{1 - \delta_0} \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \\ + \xi_0 \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^{k+1} \mathbf{k}_t + \zeta_0 \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^{k+1} \mathbf{l}_t + \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t \quad (16)$$

$$y_{j,t} = \cdots + \underbrace{\sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^{k+1}]_{(j, \cdot)} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \begin{pmatrix} \xi_0 \\ \zeta_0 \end{pmatrix}}_{\text{exogenous}_j} + \underbrace{\sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^k]_{(j, \cdot)} \cdot \boldsymbol{\varepsilon}_t}_{\text{error transformation}_j} \quad (17)$$

It is the primary econometric challenge of this section to overcome the endogeneity of $y_{j,t}$, which is clearly demonstrated in reduced-form equation (17). It contains a linear transformation of all firms’ error shocks $\boldsymbol{\varepsilon}_t$, including $\varepsilon_{i,t}$. This implies:

$$\mathbb{E}[y_{j,t} \varepsilon_{i,t}] \neq 0 \quad (18)$$

The ‘Reflection Problem’ poses a second issue as outlined in Manski (1993).

²³ $\delta_0 \geq 1$ would imply an exogenous TFP shock to firm k would increase TFP in its peer firm j more than 1x, which would increase TFP in that firm’s peer i more than 1x, and so on until the effect multiplies to infinity, or at least on the knife-edge $\delta_0 = 1$ case, any TFP shock occurring in one firm applies equally and consistently to every firm in a network. Estimating δ_0 is a helpful indicator of a multicollinear model with unstable coefficient estimates.

²⁴ Note that vectors $\mathbf{k}_t, \mathbf{l}_t$ appear twice in (16), as components of $\mathbf{y}_t$ so coefficients can be grouped but it is clearer to separate this into an ‘own-firm’ effect of $\mathbf{k}_t, \mathbf{l}_t$ and a spillover effect of $\mathbf{W}^{k+1} \mathbf{k}_t, \mathbf{W}^{k+1} \mathbf{l}_t \forall k \geq 0$

Equation (16) shows that $\mathbf{W}\mathbf{y}_t$, is a linear combination of all other regressors. In the classical setup, $\mathbf{W}$ is a block diagonal matrix of equal-size groups, and Manski (1993), later reaffirmed in Bramoullé et al. (2009) show that the reduced-form estimation of this model contains fewer estimated parameters than structural parameters, making variation in $y_{j,t}$ indistinguishable to that of all other regressors.

There are two solutions. I could either proceed with an extremely strong exogeneity assumption, (19), which is most likely satisfied by total orthogonality between all contemporaneous idiosyncratic errors hitting peer firms (20). This in turn makes the regressor $y_{j,t}$ exogenous, allowing direct estimation of (8), sidestepping the ‘Reflection Problem’ by assuming away *endogenous* regressors, and having each regressor exogenously identified in itself. For identification, this implies that any productivity shock $\varepsilon_{j,t}$ occurring in peer firm j , if it is contemporaneously correlated with TFP $z_{i,t}$ in target firm i must be attributable to the peer effect coefficient via δ_0 . This precludes any contemporaneous, correlated independent TFP shocks, which might occur on both the global and local level. Model (1.) takes this assumption in 4.2. We should assess these results with the heavy caveat that this strict error orthogonality assumption is unlikely to be credible.

$$\mathbb{E}[\varepsilon_i | \mathbf{W}^k (\boldsymbol{\alpha} \quad \gamma_t \boldsymbol{\iota} \quad \beta_0 \boldsymbol{\iota} \quad \mathbf{k}_t \quad \mathbf{l}_t)] = 0, \quad \forall k$$

$$\mathbb{E}[\varepsilon_{i,t} | \mathbf{W}^k \boldsymbol{\varepsilon}_t] = 0, \quad \forall i, j \tag{19}$$

$$\iff \mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] = 0, \quad \forall k \tag{20}$$

Instead, as recommended in Bramoullé et al. (2009), the most defensible approach is to internal instrument the *endogenous* variable, $y_{j,t}$ through deeper spatial lags of the *exogenous* parameters $\mathbf{W}^2 \mathbf{x}_t, \mathbf{W}^3 \mathbf{x}_t \dots \mathbf{W}^k \mathbf{x}_t \quad \forall k > 2$. These are valid instruments under certain testable properties of $\mathbf{W}$ given in Table I. Essentially, these reduce to checking that the matrices $\mathbb{I}, \mathbf{W}, \mathbf{W}^2$ are linearly independent, which are sufficiently achieved for group interactions by having 2 or more differently-sized groups, or a network diameter of at least 2.

This identification strategy is not a ‘free lunch’; it still requires that peer assignment, via group size, or network diameter is itself exogenously determined. This would be intuitively violated, if firms colocate endogenously, unfortunately, exactly as discussed in section 2. Some mitigation is provided by the use of time-invariant weights. Plausibly, over the observations of a panel dataset up to 20 years, if firm weights apply to each firm on a fixed basis, the temporal variation despite these constant location effects provide credible exogenous variation. On this however, the strategy runs into the limitations of Fame location data discussed in 3.3. The analysis is also helped by the relatively higher number of panel

observations to relatively low ratio of confounding movers in the dataset, as well as the numerous, plausibly exogenous reasons why a firm might choose to set up in a location with path dependency, rather than to fully maximise its peer spillover benefits. On the balance of arguments, I take the assumption of network exogeneity and identification approach for the group models in Table VI.

I use the instruments $\mathbf{W}^2\mathbf{x}_t$ and other second-order variants for this section's group models VI and $\mathbf{W}^2\mathbf{x}_t, \mathbf{W}^3\mathbf{x}_t, \mathbf{W}^4$ as feasible instruments for this section's network models in Table VII. Equation (21) illustrates the valid $2l$ moment conditions, noting that I have chosen to display this in a $l \times 2$ matrix for space and conceptual symmetry, as the instruments can be grouped by k_t and l_t regressors. The first 2 rows correspond to the exogenous regressors within the reduced-form equation itself, the own-firm classical factors effect from $k_{i,t}, l_{i,t}$ and the *exogenous* peer effect from $k_{j,t}, l_{j,t}$. The remaining rows are instrument linear combinations of all firm classical factor inputs $\mathbf{k}_t, \mathbf{l}_t$, using the higher-order spatial lags of $\mathbf{W}$ discussed.

$$\mathbb{E}[\epsilon_{i,t} \mid \begin{pmatrix} k_{i,t} & l_{i,t} \\ w_{ij}k_{j,t} & w_{ij}l_{j,t} \\ [\mathbf{W}^2]_{(j,)} \mathbf{k}_t & [\mathbf{W}^2]_{(j,)} \mathbf{l}_t \\ [\mathbf{W}^3]_{(j,)} \mathbf{k}_t & [\mathbf{W}^3]_{(j,)} \mathbf{l}_t \\ \vdots & \vdots \\ [\mathbf{W}^k]_{(j,)} \mathbf{k}_t & [\mathbf{W}^k]_{(j,)} \mathbf{l}_t \end{pmatrix}] = 0_{l \times 2} \quad (21)$$

These instruments and conditions are intuitive: we can see through equation (17) that a linear combination of all firms' contemporaneous capital and labour inputs, $\mathbf{W}^{k+1}(\mathbf{k}_t \ \mathbf{l}_t) \ \forall k \geq 0$ are an exogenous component of $y_{j,t}$ with the coefficients ξ_0 and ζ_0 respectively. It has relevance, meaning $\mathbb{E}[y_{j,t} \cdot [\mathbf{W}^k]_{(j,)}\mathbf{x}_t] \neq 0, \ \forall k > 2$ as long as $\xi_0 \neq 0$ and $\zeta_0 \neq 0$, which occurs in all unrestricted cases distinct from (11). The relevance conditions themselves are also unsurprising and highly intuitive given the model's setup with GVA equation (8). Note that the relevance conditions are more complex setup in Bramoullé et al. (2009), this is because the factor $\delta_0 \begin{pmatrix} \beta_1 & \beta_2 \end{pmatrix}$ cancels out due the accounting composition of TFP in the *exogenous effect*, which is absent from the general model. This also allows clear understanding of why the TFP-only spillover model (11) fails, as setting both $\xi_0 = \zeta_0 = 0$ means the higher order spatial lags of $\mathbf{k}_t$ and $\mathbf{l}_t$ disappears from the composition of the $y_{j,t}$ term. An Anderson-Rubin test for weak instruments in my nested 3-factor spillover setup can test for the strength of instrument of $\mathbf{k}_t$ and $\mathbf{l}_t$. If I find that the higher-order spatial lags are jointly weak instruments, that would validate the TFP-only spillover model and suggest $\xi_0 = \zeta_0 = 0$.

Finally, one might observe that $[\mathbf{W}]_{(j,)}\mathbf{x}_t$ is a component of $y_{j,t}$ and so a valid instrument. However, it does not increase the number of instruments I would use in my instrumental specification as it is already included as an exogenous regressor in 14 so does not help increase our identification of $y_{j,t}$.

4.2. LINEAR-IN-MEANS GROUP MODEL

The initial model proceeds with the strict error orthogonality (19) and (20) assumptions. This is helpful as a benchmark to compare the IV strategy, with more credible assumptions, that I then adopt. Model (1.) estimates (22), noting that the single group interactions matrix $\mathbf{W}_g$ corresponds to the *pc8* group.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t} \quad (\text{from } 9)$$

$$+ \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_0 - \delta_0 \beta_1) \cdot \sum_{j \neq i} w_{ij,g} k_{j,t} + (\zeta_0 - \delta_0 \beta_2) \cdot \sum_{j \neq i} w_{ij,g} l_{j,t}$$

$$y_{i,t} = \alpha_i + \gamma_t + \pi_0 + \pi_1 k_{i,t} + \pi_2 l_{i,t} \quad (22)$$

$$+ \pi_3 \sum_{j \neq i} w_{ij,g} y_{j,t} + \pi_4 \sum_{j \neq i} w_{ij,g} k_{j,t} + \pi_5 \sum_{j \neq i} w_{ij,g} l_{j,t} + \varepsilon_{i,t}$$

$$\mathbb{E}[\varepsilon_{i,t} \cdot (k_{i,t} \quad l_{i,t} \quad y_{j,t} \quad k_{j,t} \quad l_{j,t})'] = 0_{(5 \times 1)} \quad (23)$$

$$\begin{pmatrix} \hat{\pi}_0 \\ \hat{\pi}_1 \\ \hat{\pi}_2 \\ \hat{\pi}_3 \\ \hat{\pi}_4 \\ \hat{\pi}_5 \end{pmatrix} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \delta_0 \\ \xi_0 - \delta_0 \beta_1 \\ \zeta_0 - \delta_0 \beta_2 \end{pmatrix} \quad (24)$$

With 5 structural and 5 reduced-form parameters, this model is identified. The initial static model applies simple time-invariant weights w_{ij} in a group structure, $\mathbf{W}_{g1}$ I discussed group assignment in section 3.5.

Model (2) instruments these groups via $\mathbf{W}^2\mathbf{x}$, The reduced-form equation is given in (25). The IV strategy has the moment conditions in (21), and estimates parameters using continuously-updating Generalised Method of Moments (GMM-CUE), of calculating an efficient transformation matrix to weigh moment conditions (21) to improve the estimator's efficiency. All the IV models in this analysis are overidentified, as helpfully the structural model has 2 *exogenous* parameters in capital and labour for 1 *endogenous* GVA term. In the group model, I present additional instruments up to the 3rd-order and 4th-order lags, as the first-stage 2SLS results presented in Appendix C Table XII suggests a considerable

jump in adjusted-R² when moving from 3rd-order to 4th-order lag instruments.

$$\begin{aligned}
y_{i,t} &= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t} & (\text{from 9}) \\
&+ \delta_0 \sum_{j \neq i} w_{ij,g} y_{j,t} + (\xi_0 - \delta_0 \beta_1) \cdot \sum_{j \neq i} w_{ij,g} k_{j,t} + (\zeta_0 - \delta_0 \beta_2) \cdot \sum_{j \neq i} w_{ij,g} l_{j,t} \\
\Delta y_{i,t} &= \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} & (25) \\
&+ \beta_3 \Delta \sum_{j \neq i} w_{ij,g} y_{j,t} + \beta_4 \Delta \sum_{j \neq i} w_{ij,g} k_{j,t} + \beta_5 \Delta \sum_{j \neq i} w_{ij,g} l_{j,t} + \Delta \varepsilon_{i,t} \\
\mathbb{E}[\epsilon_{i,t} \mid &\begin{pmatrix} k_{i,t} & l_{i,t} \\ w_{ij} k_{j,t} & w_{ij} l_{j,t} \\ \begin{bmatrix} \mathbf{W}_g^2 \end{bmatrix}_{(j,)} \mathbf{k}_t & \begin{bmatrix} \mathbf{W}_g^2 \end{bmatrix}_{(j,)} \mathbf{l}_t \\ \begin{bmatrix} \mathbf{W}_g^3 \end{bmatrix}_{(j,)} \mathbf{k}_t & \begin{bmatrix} \mathbf{W}_g^3 \end{bmatrix}_{(j,)} \mathbf{l}_t \\ \vdots & \vdots \\ \begin{bmatrix} \mathbf{W}_g^k \end{bmatrix}_{(j,)} \mathbf{k}_t & \begin{bmatrix} \mathbf{W}_g^k \end{bmatrix}_{(j,)} \mathbf{l}_t \end{pmatrix}] = 0_{l \times 2} & (\text{from 21})
\end{aligned}$$

Results for Model (1.) and (2.) are given in Table VI.

As a brief extension, I explore regressing multiple, non-overlapping spatial groups at once. This is a natural examination of the specification as I already have the $\mathbf{W}_{g1}$, $\mathbf{W}_{g2}$, $\mathbf{W}_{g3}$ concentric doughnut group transformations for *pc8*, *pc4* and *ttwa* to run all in the panel IV model at once. The algebraic derivation of endogeneity and valid instruments is very similar to those in the single-regressor problem, where the inversion term is the sum matrices weighted by their coefficient δ_s .

TABLE VI
LMM: 1 peer regressor with group weights, error
orthogonality model vs instrumented

	(1.) exog	(2.) reduced-iv	(3.) iv4
β_1	0.261 (0.005)***	0.233 (0.008)***	0.234 (0.008)***
β_2	0.650 (0.005)***	0.555 (0.007)***	0.555 (0.007)***
δ_0	0.199 (0.004)***	-0.384 (0.177)**	-0.414 (0.177)**
ξ_0	-0.005 (0.002)**	0.029 (0.014)**	0.031 (0.014)**
ζ_0	0.011 (0.003)***	0.035 (0.013)***	0.037 (0.013)***
J-test		X (0.330)	X (0.357)
Instr.		$\mathbf{W}_{g1}^2 \mathbf{k}, \mathbf{W}_{g1}^2 \mathbf{l}$ $\mathbf{W}_{g1}^3 \mathbf{k}, \mathbf{W}_{g1}^3 \mathbf{l}$	$\mathbf{W}_{g1}^2 \mathbf{k}, \mathbf{W}_{g1}^2 \mathbf{l}$ $\mathbf{W}_{g1}^3 \mathbf{k}, \mathbf{W}_{g1}^3 \mathbf{l}$ $\mathbf{W}_{g1}^4 \mathbf{k}, \mathbf{W}_{g1}^4 \mathbf{l}$
i F.E.	✓	✓	✓
t F.E.	✓	✓	✓
Obs.	726,283	587,772	587,772
R^2 (excl. F.E.)	0.3550	0.0012	-0.0139
R^2 (Overall)	0.7966	0.0000	0.0000

Regression tables given structural parameters calculated using transformations of the reduced-form parameters. Stars denote significance against an H_0 with the usual confidence levels, but these are not meaningful for this analysis due to causal interpretations discussed.

$$\mathbf{y}_t = \boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \quad (26)$$

$$\begin{aligned} & + \sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \mathbf{y}_t + \sum_{s=1}^3 (\xi_s - \delta_s \beta_1) \mathbf{W}_{gs} \mathbf{k}_t + \sum_{s=1}^3 (\zeta_s - \delta_s \beta_2) \mathbf{W}_{gs} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \\ & = (\mathbb{I} - \sum_{s=1}^3 \delta_s \mathbf{W}_{gs})^{-1} \left(\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \boldsymbol{\varepsilon}_t \right. \\ & \quad \left. + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t \right) \\ & = \sum_{k=0}^{\infty} \left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k \boldsymbol{\alpha} + \frac{\gamma_t + \beta_0}{1 - \sum_{s=1}^3 \delta_s} \boldsymbol{\iota} + \sum_{k=0}^{\infty} \left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k (\beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t) \quad (27) \\ & \quad + \sum_{k=0}^{\infty} \left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k \sum_{s=1}^3 \mathbf{W}_{gs} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \begin{pmatrix} \xi_s - \delta_s \beta_1 \\ \zeta_s - \delta_s \beta_2 \end{pmatrix} + \sum_{k=0}^{\infty} \left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k \boldsymbol{\varepsilon}_t \end{aligned}$$

$$\begin{aligned} y_{j,t} = & \cdots + \underbrace{\sum_{k=0}^{\infty} \left[\left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k \sum_{s=1}^3 \mathbf{W}_{gs} \right]_{(j,)} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \begin{pmatrix} \xi_s - \delta_s \beta_1 \\ \zeta_s - \delta_s \beta_2 \end{pmatrix}}_{\text{exogenous}_j} \\ & + \sum_{k=0}^{\infty} \left[\left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k \right]_{(j,)} \cdot \boldsymbol{\varepsilon}_t \end{aligned} \quad (28)$$

This strategy of multiple weighting groups at once opens up interesting additional cross-transformation instruments. These cross-transformations are possible and create new, linearly independent regressors as long as there is sufficient heterogeneity between the weighting matrices, which here is theoretically guaranteed by the mutual orthogonality of $\mathbf{W}_g$. Any combination of higher-order $\mathbf{W}_{gs}$ group transformations can be employed as instruments (29), or the linear composite of the 3 group transformation matrices, which intuitively represents all firms in a *ttwa* can be used in higher orders (30).

$$\mathbb{E}[\epsilon_{i,t} \mid \begin{pmatrix} [\mathbf{W}_{g1} \mathbf{W}_{g2}]_{(j,)} \mathbf{k}_t & [\mathbf{W}_{g1} \mathbf{W}_{g2}]_{(j,)} \mathbf{l}_t \\ [\mathbf{W}_{g2} \mathbf{W}_{g3}]_{(j,)} \mathbf{k}_t & [\mathbf{W}_{g2} \mathbf{W}_{g3}]_{(j,)} \mathbf{l}_t \\ \vdots & \vdots \end{pmatrix}] = 0 \quad (29)$$

$$\mathbb{E}[\epsilon_{i,t} \mid \begin{pmatrix} \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g1} \right]_{(j,)} \mathbf{k}_t & \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g1} \right]_{(j,)} \mathbf{l}_t \\ \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g2} \right]_{(j,)} \mathbf{k}_t & \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g2} \right]_{(j,)} \mathbf{l}_t \\ \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g3} \right]_{(j,)} \mathbf{k}_t & \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g3} \right]_{(j,)} \mathbf{l}_t \end{pmatrix}] = 0_{3 \times 2} \quad (30)$$

Importantly, the reduced form in (28) shows that the coefficients on the *exogenous effect* regressors are no longer merely ξ_0, ζ_0 , but the linear combination

given. This model therefore provides a theoretical route to identifying and testing (11), the z-spillover model, should that be the researcher’s preferred specification.

However, in practice, regressing more groups proves unlikely to offer more insights with this dataset and holistic approach without additional diagnosis. Coefficients reported in Appendix C Table XIII appear unstable, though this is slightly alleviated by omitting the outer *ttwa* group and regressing only using *pc8* and *pc4* $\mathbf{W}_g$ transformations. Nevertheless, in the results given here, these fail the standard Sargan-Hansen test for overidentifying restrictions. In addition, employing more instruments, the higher-order cross-transformations of $\mathbf{W}_{gs}$ discussed is theoretically sound, but led to multicollinearity in this dataset through any approach. This could be due to the fact that each $\mathbf{W}_{gs}$ is not truly independent and orthogonal, the identities of *pc8* and *pc4* are linked.

This extension is interesting and useful because the assignment of multiple groups, especially through more sophisticated dimensions as discussed in section B.1 through industry and regional cuts, could lead to better understanding of firm’s heterogenous peer groups. Perhaps these alternative group choices will have more success by producing independent, orthogonal matrices though they run beyond the scope of the simple spatial group identities selected here.

4.3. STATIC DISTANCE DECAY

I employ a similar static specification strategy, but employing $\mathbf{W}_d$ distance-decayed weights in a network structure, rather than group weights. The reciprocal distance weights are stylistic of a declining peer influence further away, representing a lower frequency of social collisions that might inspire knowledge spillovers, barriers created via transport and time costs, and decreased ease of peer observation. Similarly, if this analysis' hypothesis is that spatial proximity is key to determination of a peer group, it is intuitive that increasing distance, as a continuous measure, would imply a decline in influence, rather than equal influence of all peers in an arbitrary cut-off.

I estimate this model employing the spatial lag instruments only, dropping the error orthogonality variant. The reduced-form equation and moment conditions of these models are identical to those of the group setting, except for the use of different weights which show up in $w_{ij,d}$ and $\mathbf{W}_d$.

$$\begin{aligned}
y_{i,t} &= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t} & (\text{from 9}) \\
&+ \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_0 - \delta_0 \beta_1) \cdot \sum_{j \neq i} w_{ij,g} k_{j,t} + (\zeta_0 - \delta_0 \beta_2) \cdot \sum_{j \neq i} w_{ij,g} l_{j,t} \\
\Delta y_{i,t} &= \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} & (31) \\
&+ \beta_3 \Delta \sum_{j \neq i} w_{ij,d} y_{j,t} + \beta_4 \Delta \sum_{j \neq i} w_{ij,d} k_{j,t} + \beta_5 \Delta \sum_{j \neq i} w_{ij,d} l_{j,t} + \Delta \varepsilon_{i,t} \\
\mathbb{E}[\epsilon_{i,t} \mid &\begin{pmatrix} k_{i,t} & l_{i,t} \\ w_{ij} k_{j,t} & w_{ij,d} l_{j,t} \\ [\mathbf{W}_d^2]_{(j,)} \mathbf{k}_t & [\mathbf{W}_d^2]_{(j,)} \mathbf{l}_t \\ [\mathbf{W}_d^3]_{(j,)} \mathbf{k}_t & [\mathbf{W}_d^3]_{(j,)} \mathbf{l}_t \\ \vdots & \vdots \\ [\mathbf{W}_d^k]_{(j,)} \mathbf{k}_t & [\mathbf{W}_d^k]_{(j,)} \mathbf{l}_t \end{pmatrix}] = 0_{l \times 2} & (\text{from 21})
\end{aligned}$$

Results for each of the 3 reciprocal distance measures are given in Table VII. Note that each of the distance d_1, d_2, d_3 measures are distinct transformations and so the properties discussed for the W_{gs} matrices allowing them to be regressed at once do not here apply. In addition, I include a specification exploring employing more lags for overidentification.

Model (4.) includes more lags which improves significance. The inclusion of more lags continues to pass the Hansen J-test, avoiding over-specification which would be tested by a rejection of the null hypothesis. I include an additional table in Appendix D which shows the first-stage Panel OLS estimates of the endogenous regressor $\mathbf{W}_y$ using the instruments employed, were this specification

TABLE VII
Static panel IV using continuous distance-decay $\mathbf{W}_d$

	(1.) dd1	(2.) dd2	(3.) dd3	(4.) dd1-deep3
β_1	0.271 (0.007)***	0.271 (0.007)***	0.271 (0.007)***	0.267 (0.007)***
β_2	0.512 (0.006)***	0.512 (0.006)***	0.512 (0.006)***	0.513 (0.006)***
δ_0	-0.159 (0.286)	-0.110 (0.233)	-0.353 (0.309)	-0.740 (0.388)*
ξ_0	0.011 (0.019)	0.008 (0.017)	0.033 (0.030)	0.052 (0.029)*
ζ_0	0.023 (0.037)	0.015 (0.027)	0.049 (0.041)	0.098 (0.050)*
J-test	X (0.311)	X (0.989)	X (0.969)	X (0.100)
Instr.	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$	$\mathbf{W}_{d2}^2 \mathbf{k}, \mathbf{W}_{d2}^2 \mathbf{l}$	$\mathbf{W}_{d3}^2 \mathbf{k}, \mathbf{W}_{d3}^2 \mathbf{l}$	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$ $\mathbf{W}_{d1}^3 \mathbf{k}, \mathbf{W}_{d1}^3 \mathbf{l}$
i F.E.	✓	✓	✓	✓
t F.E.	✓	✓	✓	✓
Obs.	854,192	854,192	854,192	854,192
R^2	0.0985	0.1034	0.0980	-0.1213

to be estimated with 2SLS²⁵. This shows improvements in adjusted R^2 for the model up to 3 lags, but barely any improvement for including an additional 4th-order lag.

²⁵ Noting that in practice, I use GMM-CUE for efficiency of estimates

To investigate this issue, in Appendix D Table XV I run an extension where firms which share a location, meaning distance is mechanically 0, are excluded. Results are reported in XV. On evaluation by R^2 these models perform slightly worse. An odd phenomenon is that the signs flip on the spillover parameters. These specifications reveal an emerging feature of these models, which is that $\delta_0 < 0 \implies \xi_0 > 0 \cap \zeta_0 > 0$ and vice-versa. The sign flipping in these sets of models suggests some joint parameter exists as a linear combination of the parameters I have estimated, which these models are trying to fit.

5. DYNAMIC MODELS

The previous section’s static models requires a strict factor exogeneity assumption outlined in equation (12) for all firms, including the own-firm effect of the production function. In this section, I use a dynamic model to relax this assumption. I also introduce a vector of temporal regressors: an autoregressor of own-firm GVA as a component of its TFP, and allowing for the past peer TFP components I analysed in the static model to influence current-period TFP in a lagged fashion. The structure of this specification now resembles a vector panel, which I will denote with $\text{VAR}(p,q)$ ²⁶.

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \sum_{k=1}^p \phi_k y_{i,t-k} + \sum_{k=0}^q \mathbf{z}_{t-k}^{\text{spill}} \boldsymbol{\theta}_k + \varepsilon_{i,t} \quad (32)$$

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \sum_{k=1}^p \phi_k y_{i,t-k} + \varepsilon_{i,t} \quad (33)$$

$$+ \sum_{k=0}^q \left[\delta_k \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_k - \delta_k \beta_1) \cdot \sum_{j \neq i} w_{ij} k_{j,t} + (\zeta_k - \delta_k \beta_2) \cdot \sum_{j \neq i} w_{ij} l_{j,t} \right]$$

Adding the autoregressor $y_{i,t}$ is highly desirable. The panel setup with firm fixed effects already accounts for own-firm factors, and the presentation of R^2 including and excluding variation accounted for by the fixed effects throughout this paper demonstrates this. In the peer effects setting, autoregressive lags may also help control for reverse causality. If $y_{i,t-1}$ is causal for z_{jt} , or feasibly $z_{j,t-1}$ and $\mathbb{E}[z_{j,t-1}, z_{j,t}] \neq 0$, then the inclusion of $y_{i,t-k}$ eliminates that bias. I offer no *a priori* view on the strength of ϕ_1 , the autoregressor coefficient. A significant estimate would imply heavy bias to the static models, validating the use in this dynamic setting.

There are also significant theoretical advantages to introducing the q -lags of the spatial spillover. This specification with both spatial and temporal lags accounts for the possibility that the spillover effect from firm j ’s production factor has a lagged impact on i , conceptualisable as the formation of clusters from firms seeing the benefits of colocation over some time after a productivity shock. Without including the q lags within $z_{i,t}$, if $\mathbf{z}_{t-k}^{\text{spill}}$ is correlated with $\mathbf{z}_t$, then estimating δ_0 in a static model when the true effect is dynamic exposes estimation to omitted variable bias (OVB) by attribution of the lagged to the current period’s regressor.

In practice for this dataset however, it is unlikely that more-than-one period spillover lag is relevant. This is due to this analysis using Fame’s annual frequency

²⁶ q -lags refers to those on the peer spillover effect, rather than those of a moving average component as is traditional of a VARMA(p,q) specification

observations. As a result, I show the structural equations in an AR(1,0) structure so that the primary advancement on the static models are purely the use of the weak exogeneity assumption, and first-order autoregressor.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \phi_1 y_{i,t-k} + \varepsilon_{i,t} \quad (34)$$

$$+ \delta_k \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_0 - \delta_0 \beta_1) \cdot \sum_{j \neq i} w_{ij} k_{j,t} + (\zeta_0 - \delta_0 \beta_2) \cdot \sum_{j \neq i} w_{ij} l_{j,t}$$

Estimation of this model is near-identical in reduced form to the coefficients described in the static problem, requiring only that I estimate one additional parameter $\hat{\phi}_1$. Instead, the crucial change comes in the choice of instruments. The dynamic specification allows the investigation to employ and contrast the use of temporal higher-order lags, rather than spatial ones.

5.1. DYNAMIC PANEL IDENTIFICATION

The dynamic panel specification uses a autocorrelation error orthogonality assumption to find valid identification (35). This replaces (12) and instead allows for some moderate level of cross-sectional error correlation (36).

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{i,t-1}] = 0, \quad \forall t, t-1 \quad (35)$$

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] \neq 0, \quad \exists i, j \quad (36)$$

$$\mathbb{E}[\varepsilon_t] = 0, \mathbb{E}[\varepsilon_t^2] = \sigma_t^2, \mathbb{E}[\varepsilon_t | \Omega_{t-1}] = 0, \quad \forall i \quad (\text{w.n.}) \quad (37)$$

No autocorrelation of errors may generally be seen as a more credible assumption than strictly no cross-correlation. In a dynamic, correctly-specified model with the correct autoregressive properties, time autocorrelation as component of the error term are modelled out. Conceptually, we could also argue that firms make efficient decisions, so if there is a shock observed in this period, firms ensure that in the next, any expected factor inputs that are realised by that period fully take advantage of the shock and its effect is then captured by the input regressors. Any remaining residual must have been expected and so mean zero (formally a Martingale sequence). Further econometric analyses could guarantee this assumption by plotting patterns of errors and check for autocorrelation, which could come from moving average or omitted latent variable structures to the specification missing here. This specification can be identified in reduced panel form, which requires first differences to eliminate fixed effects, allowing me to retain the use of firm fixed effects.

Estimation under these dynamic panel conditions are developed by Anderson and Hsiao (1981), with further techniques expanded in Arellano and Bond (1991)

and Blundell and Bond (1998) which improve efficiency. The estimator given in Arellano and Bond (1991) is designed to allow for endogeneity, limiting only to weak exogeneity as I do here. Internal instruments of the endogenous regressors are found using their lagged first differences of the log variables. Identification is improved in the System GMM approach given in Blundell and Bond (1998) by also including the first lagged, not differenced, lag of the endogenous regressors which meets the usual instrument conditions, as an instrument. I estimate this section's results with System GMM²⁷. Valid instruments are therefore theoretically derived as $\Delta x_{i,\tau-j}$ for any j if $x_{i,\tau}$ is a valid instrument.

5.2. EXPLORING TEMPORAL LAG STRUCTURES

I firstly relax the strict exogeneity assumption of the standard factors to weak exogeneity. There are two approaches to this: capital and labour can be assumed to be predetermined: capital faces some time-to-build constraint where each firm allocates investment in the previous period based on the limited information available then, Ω_{t-1} . The predetermined nature of capital is a common modelling choice of firm production in a dynamic system, though usually with quarterly observations. A similar sticky decision pattern is assumed of labour, perhaps through job searching and hiring frictions. For handling panel data in an autoregressive setting, first-differencing will lead to the reduced-form error term $\Delta \varepsilon_t$ spanning 2 periods. As a result, weak exogeneity of the transformed factors despite the factors being predetermined in levels in period t requires lags of $t - 1$. These lags are valid instruments with defensible exclusion restrictions. Since the zero autocorrelation property applies to any firm i , the full vector of deep lag factor regressors and their transformations are plausibly exogenous.

Pre-determined factors:

$$K_{i,t} = I_{i,t} + dK_{i,t-1}$$

$$I_{i,t} = \mathbb{E}[I_{i,t}^* | \Omega_{t-1}]$$

$$L_{i,t} = \mathbb{E}[L_{i,t}^* | \Omega_{t-1}]$$

$$K_{i,t}, L_{i,t} \in \Omega_{t-1}$$

$$\mathbb{E}[\varepsilon_{i,t} | \Omega_{t-1}] = 0$$

$$\mathbb{E}[\varepsilon_{i,t} | (k_{i,\tau} \quad l_{\tau})] = 0_{(1 \times t)}, \quad \forall i \quad \forall \tau \leq t \quad (38)$$

$$\mathbb{E}[\Delta \varepsilon_{i,t} | (\Delta \mathbf{W}^k \mathbf{k}_{\tau-1} \quad \Delta \mathbf{W}^k \mathbf{l}_{\tau-1})] = 0_{(1 \times t)}, \quad \forall k \quad \forall \tau \leq t \quad (39)$$

Alternatively, capital and labour may be flexible inputs: at least partially

²⁷ Implemented with Dazhong Wu and Jian Hua and Feng Xu (2023), with heavy modifications in Aloysius Lip (2026)

able to react to shocks as part of the decision set at time t . This may be a more realistic assumption when handling annual data, such as Fame: shocks occurring at the beginning of a calendar year would have no arguable reason why the firm's input choices later in that year hadn't reacted to knowledge of the shock, any more than it had occurred at the end of the previous year. It is therefore much harder to argue the plausibly exogeneity of contemporaneous factors to the error term. In practice, as with any dynamic panel model, the procedure laid out in this paper is not bound by the features of this particular dataset, which happens to be annual. Future analysis repeating this question with quarterly observations may well exploit the plausible predetermined nature of some production factors. In these cases, exogeneity in the reduced form first-differenced model require deep lags of $t - 2$ for exogeneity.

Flexible factors:

$$K_{i,t}, L_{i,t} \in \Omega_t$$

$$\mathbb{E}[\varepsilon_{i,t} | \Omega_{t-1}] = 0$$

$$\mathbb{E}[\varepsilon_{i,t} | (k_{i,\tau-1} \quad l_{\tau-1})] = 0_{(1 \times t)}, \quad \forall i \quad \forall \tau \leq t \quad (40)$$

$$\mathbb{E}[\Delta \varepsilon_{i,t} | (\Delta \mathbf{W}^k \mathbf{k}_{\tau-2} \quad \Delta \mathbf{W}^k \mathbf{l}_{\tau-2})] = 0_{(1 \times t)}, \quad \forall k \quad \forall \tau \leq t \quad (41)$$

It is certain that the autoregressor, y_{t-1} is endogenous. y_{t-1} mechanically contains a ε_{t-1} residual term; first differencing for the reduced-form panel means that $\Delta \varepsilon_t$ and Δy_t will contain a linear combination of ε_t and ε_{t-1} . When Δy_{t-1} contains a linear combination of ε_{t-1} and ε_{t-2} , they share a common component, violating exogeneity. Here, a simple first-order lag is sufficient temporal distance to give an instrument with exclusion.

$$\Delta y_{i,t-2} \propto \Delta \varepsilon_{i,t-2} \Delta \varepsilon_{i,t-3} \quad (42)$$

$$\mathbb{E}[\Delta \varepsilon_{i,t} | (\Delta \varepsilon_{i,t-2} \quad \Delta \varepsilon_{i,t-3})] = 0_{1 \times 2} \quad (43)$$

This is additionally confirmed by Anderson-Rubin (AR) tests across all specifications, which testing the autoregressor Δy_{t-1} , consistently reject the null hypothesis at 1% significance. By contrast, the AR(2) test, which checks for the same weak instruments but at the second-order lag Δy_{t-2} , there is repeatedly insufficient evidence to reject the null hypothesis, again confirming the theory of the System GMM that deep lags are a valid instrument. With coefficients estimated across such a vast and heterogenous dataset, these consistent results for the basic properties are useful to observe.

As the properties of the spillover regressor, $\mathbf{z}^{\text{spill}}$ are unknown *a priori*, I employ a $t - 2$ deep lag at minimum to instrument for that component. This

leaves only the question about the classical regressors as discussed.

In this dataset, the empirical experiments run suggest that the decision over whether the classical factors should be predetermined or not make no difference.

Table VIII presents results from the reduced model (34), varying whether capital alone is a predetermined factor, meaning its lags can start as soon as the $t - 1$ period in Model (1.), whether labour is alone (2.), or they are both endogenous, requiring both their instrument lags to start from the $t - 2$ period.

Across all specifications, the test-statistic for the Sargan-Hansen J-test is insignificant, suggesting insufficient evidence that the exogenous moment condition can be met, requiring any change to the predetermined assumptions. This offers 2 pathways: either I can use the results of this test with confidence and proceed with the assumption of capital and labour as predetermined variables. This offers an additional time-axis observation, and thereby degree of freedom, to conduct this panel analysis. Alternatively, I can be cautious and continue to employ deep lags of at least $t - 2$ for capital and labour instruments. I proceed with the latter for this section's exposition, owing to the richness of Fame in the time dimension. Data samples with alternative cuts, such as subsections of the data examining a certain time, industry, or region, may justify the alternative option at this stage.

TABLE VIII
Dynamic panel: baseline instrumental specification

	(1.) k-pred	(2.) l-pred	(3.) both-endog
β_1	0.341 (0.123)	0.300 (0.150)	0.352 (0.145)
β_2	0.700 (0.249)	0.645 (0.151)	0.510 (0.237)
ϕ_1	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
δ_0	-0.076 (0.390)	-0.150 (0.335)	0.195 (0.390)
ξ_0	-0.193 (0.239)	0.025 (0.237)	-0.043 (0.278)
ζ_0	0.511 (0.489)	0.162 (0.418)	0.493 (0.391)
J-test	X (0.834)	X (0.690)	X (0.913)
AR(1) test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) test	X (0.129)	X (0.154)	X (0.294)
Instr.	$\mathbf{y}_{t-2:3}, \mathbf{k}_{t-1:2}$ $\mathbf{l}_{t-2:3}, \mathbf{W}_{d1}\mathbf{y}_{t-2:3}$ $\mathbf{W}_{d1}\mathbf{k}_{t-1:2}, \mathbf{W}_{d1}\mathbf{l}_{t-2:3}$	$\mathbf{y}_{t-2:3}, \mathbf{k}_{t-2:3}$ $\mathbf{l}_{t-1:2}, \mathbf{W}_{d1}\mathbf{y}_{t-2:3}$ $\mathbf{W}_{d1}\mathbf{k}_{t-1:2}, \mathbf{W}_{d1}\mathbf{l}_{t-2:3}$	$\mathbf{y}_{t-2:3}, \mathbf{k}_{t-2:3}$ $\mathbf{l}_{t-2:3}, \mathbf{W}_{d1}\mathbf{y}_{t-2:3}$ $\mathbf{W}_{d1}\mathbf{k}_{t-2:3}, \mathbf{W}_{d1}\mathbf{l}_{t-2:3}$
i F.E.	✓	✓	✓
t F.E.	✓	✓	✓
Obs.	717,300	717,300	717,300

Hansen J-test for over-identification: ‘X’ indicates insufficient evidence at the 95% level to reject the null-hypothesis that the model is identified.

Anderson-Rubin AR(1) test: ‘✓’ indicates the use of single-order lags is a weak instrument, as expected from a first-differencing System GMM approach.

Anderson-Rubin AR(2) test: ‘X’ indicates insufficient evidence to reject the use of second-order lags due to their being a weak instrument, as expected from the first-differencing with lags greater than 2nd-order to overcome contemporaneous endogeneity. ‘checkmark’ indicates some instruments used were weak.

5.3. HIGHER-ORDER INSTRUMENTS

A decomposition of the spatially endogenous component of the regression, $\mathbf{W}y_t$, has a similar form to the static problem. It is helpful to show this to aid the construction of valid instruments: it is obvious writing $y_{j,t}$ in reduced form in (??) from where an exogenous instrument might come.

$$\mathbf{y}_t = \boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \phi_1 \mathbf{y}_{t-1} + \delta_0 \mathbf{W} \mathbf{y}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \quad (44)$$

$$\begin{aligned} &= \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \left(\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \phi_1 \mathbf{y}_{t-1} \right. \\ &\quad \left. + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \right) \\ y_{j,t} &= \cdots + \underbrace{\phi_1 y_{j,t-1}}_{\text{untransformed lag}_j} + \underbrace{\sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^{k+1}]_{(j,)} (\mathbf{k}_t \ \mathbf{l}_t) \begin{pmatrix} \xi_0 \\ \zeta_0 \end{pmatrix}}_{\text{contemporaneous exogenous}_j} \\ &\quad + \underbrace{\sum_{k=0}^{\infty} [\phi_1 \delta_0^k \mathbf{W}^k]_{(j,)} \mathbf{y}_{t-1}}_{\text{lagged spatial}_j} + \underbrace{\sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^k]_{(j,)} \cdot \boldsymbol{\varepsilon}_t}_{\text{error transformation}_j} \end{aligned} \quad (45)$$

The comparison to equation (17) is evident: the contemporaneous spatial lag term is still present, meaning $[\mathbf{W}^k]_j, (\mathbf{k}_t \ \mathbf{l}_t)$ is still a valid instrument. However, this may not be so useful under a cautious approach allowing for cross-sectional non-orthogonality of errors (36), as $\mathbf{W}_j, (\mathbf{k}_t \ \mathbf{l}_t)$, the spatial *exogenous effect* regressor is itself endogenous, or only weakly exogenous, so higher-order spatial and temporal lags should be used to instrument for this term already.

This decomposition yields several over new exogenous terms due to the lags involved. The first is the simple untransformed lag component, $\phi_1 y_{j,t-1}$, which still persists in the reduced form even after the spatial lag inversion. This permits the use of simple deep lags, $y_{j,t-2}, y_{j,t-3}, \dots$ as instruments. There is an additional combined temporal and spatial lag transformation component, $[\mathbf{W}^k]_j, y_{t-1}$. This derivation shows that there are up to 3 functionally unique variable transformations which can be used as a valid instrument in this analysis: I estimate the equation fulfilling the vector of moment conditions, (46) via GMM.

$$\mathbb{E}[\Delta \varepsilon_{it} | \begin{pmatrix} \Delta y_{j,t-2} \\ \Delta [\mathbf{W}^k]_{(j,)} y_{t-\tau} \\ \Delta [\mathbf{W}^k]_{(j,)} k_{t-\tau} \\ \Delta [\mathbf{W}^k]_{(j,)} l_{t-\tau} \end{pmatrix}] = 0 \forall k \forall \tau \geq 2 \quad (46)$$

It is a function of this analysis' unique application of a dynamic panel GMM to a peer effects model that offers this plethora of lags and instruments to employ and credibly eliminate exogeneity from the peer effects *endogenous* regressor while gaining an impressive amount of variance.

I firstly adopt the classic dynamic panel GMM strategy proposed by Anderson and Hsiao (1981) and simply use the deeper lags of $y_{j,t-2}, \dots$ as instruments. With a rich temporal dataset such as Fame, multiple deep lags allows for over-identification, provided that additional information is being gained. I leave it for future analysis to employ Information Criteria tests to determine optimal number of lags in the simple autoregressor instrument. In Table IX, I simply estimate (34) varying the number of autoregressor lags from 3 lags between $t - 2$ to $t - 4$, to 5 lags between $t - 2$ to $t - 6$. While these results don't offer any obvious new

TABLE IX
Dynamic panel: deeper instrumental lags

	(1.) deep-4	(2.) deep-5	(3.) deep-6
β_1	0.241 (0.112)	0.171 (0.101)	0.212 (0.087)
β_2	0.725 (0.175)	0.783 (0.161)	0.701 (0.149)
δ_0	-0.004 (0.354)	0.123 (0.326)	0.150 (0.296)
ϕ_1	-0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
ξ_0	0.277 (0.254)	0.290 (0.234)	0.097 (0.191)
ζ_0	0.007 (0.327)	-0.155 (0.275)	0.063 (0.237)
J-test	X (0.456)	X (0.664)	X (0.391)
AR(1) test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) test	X (0.138)	X (0.142)	X (0.228)
Instr.	$\mathbf{y}_{t-2:4}, \mathbf{k}_{t-2:4}$ $\mathbf{l}_{t-2:4}, \mathbf{W}_{d1}\mathbf{y}_{t-2:4}$ $\mathbf{W}_{d1}\mathbf{k}_{t-2:4}, \mathbf{W}_{d1}\mathbf{l}_{t-2:4}$	$\mathbf{y}_{t-2:5}, \mathbf{k}_{t-2:5}$ $\mathbf{l}_{t-2:5}, \mathbf{W}_{d1}\mathbf{y}_{t-2:5}$ $\mathbf{W}_{d1}\mathbf{k}_{t-2:5}, \mathbf{W}_{d1}\mathbf{l}_{t-2:5}$	$\mathbf{y}_{t-2:6}, \mathbf{k}_{t-2:6}$ $\mathbf{l}_{t-2:6}, \mathbf{W}_{d1}\mathbf{y}_{t-2:6}$ $\mathbf{W}_{d1}\mathbf{k}_{t-2:6}, \mathbf{W}_{d1}\mathbf{l}_{t-2:6}$
i F.E.	✓	✓	✓
t F.E.	✓	✓	✓
Obs.	717,300	717,300	717,300

insight from parameter estimates, they do confirm that continuing with the higher simple autoregressor lag strategy does not leave the estimation over-identified,

with a continuing failure to reject the Sargan-Hansen J-test’s null hypothesis.

Due to the time constraints of this investigation, I leave it for future analysis to continue the narrative and exploration here to continue to test each type of instrument individually and in combination with each other. As I deployed in Static models, estimating the first-stage of the reduced-form equation with simple OLS and comparing R^2 would offer insights on the strengths of instruments employed to explain the variation in the endogenous regressors. Additionally, as already discussed, Information Criteria tests would also helpfully determine optimal lags for use in instruments. My hypothesis is some combination of the untransformed autoregressive deep lag and spatially-transformed lagged *endogenous* regressor would significantly improve identification.

5.4. DYNAMIC EMPIRICAL RESULTS

The empirical tests so far as given in Table VIII and Table IX offer very little support for the deployment of the dynamic structure, with a consistently zero parameter on the autoregressor coefficient ϕ_1 . Additionally, specifications run so far with AR(1,q) structures of lag on the spatial components so far fail to find any significant coefficients for $\theta_1, \theta_2, \dots$ as occurs in the static case. Taken as face value, this offers some credibility to the use of a static model to outline the problem. Especially with annually-frequency panel data, results from this section would argue that nearly all the variation of interest is explainable by the contemporaneous classical and spatial regressors employed in section 4.

Taken seriously, this would suggest that spatial productivity spillovers are a static problem. As a policy question, it implies that firms are rational, efficient actors who take advantage or colocate in an area to take advantage of the contemporaneous spillovers exhibited from their high-performing neighbours. The dynamic formation of a cluster either therefore reflect repeated, serially uncorrelated productivity shocks in a region attributable to other factors than merely the spillover between high-performing firms, or that firms move to a cluster continually over time for some other driver than chasing the highest spillover to their TFP levels.

Overall, this section provides a framework to continuing to test the use of dynamic instruments which may improve the explanatory power of lagged components of the dynamic specification, reaffirms theory from Arellano and Bond (1991) on the use of deep lags as valid instruments, and further identifies interesting combinations of spatial and temporal lags that could prove powerful relevance terms to improve identification.

6. DISCUSSION AND EXTENSIONS

This analysis achieves 3 of the 4 purposes it set out in section 1 well. I develop robust data science techniques appropriate for the task that allow efficient handling of mass spatial data of any size required for a credible spatial analysis, for which Fame likely represents the upper bound of observations. I apply the econometric theory of deploying higher-order spatial lags in a contemporaneous peer effect model, and demonstrates that this tool is useful in the critical setting of analysing spatial productivity. In this, I offer key insights on the contribution and importance of different specifications varying weights of spatial peers, and on the employment of different weighting structures, such as multiple non-overlapping groups, to improve identification.

I extend this to the dynamic setting: I show that alternative deeper lags, this time on the temporal dimension, are permissible, aligning theory presented on dynamic panel structures from Arellano and Bond (1991) and estimates proposed by Blundell and Bond (1998) to a peer effects application, again, particularly that the policy-crucial question of spatial productivity can be investigated through these tools. I offer insights on the complimentary and overlapping nature of spatial and temporal lags, establishing a framework of investigation to evaluate the performance of selection of instruments. Additionally, I find some early evidence supporting the use of a static model in this question, with minimal significant results for dynamic effects for the spatial productivity spillover. Taken seriously, this provokes re-examination of the centrifugal forces occurring over time, inspiring colocation and urban development, if it implies that firms believe the benefits of positive productivity shocks to be durable.

Nevertheless, the last purpose listed, aiming to achieve robust empirical estimates of the spillover are inconsistent. *A priori* expectations of the key parameters estimated, δ_0, ξ_0, ζ_0 would expect them to be positively signed. A negative sign on some taken as truth, would imply competing, centrifugal effects such as congestion or constrained input resources. Even then, results presented are unusual for finding negative signs on some coefficients and not others, indeed specifications seem to consistently imply that $\delta_0 > 0 \implies \xi_0 < 0, \zeta_0 < 0$, and vice versa. This is an unusual pattern that most likely suggests misspecification: models are most likely trying to fit around some combination of the parameters of interest where the factors compete for contribution, and a sudden attribution to one via random noise signifies a decrease to the other. Alternatively, the unusual pattern of parameter movement could be some accounting

This paper handles several complicated econometric issues that require significant further scrutiny to yield a definitive answer to the nature of these results.

Any one of the uncertain model choices discussed in this paper, likely in order of significance, the arbitrary deselection of peers by a 5km distance boundary, the choice to only construct distance between peers by spatial distance, could have been the key to invalid peer identity. Further econometric investigation is required those which are most key. From this, robust, policy-relevant econometric estimates can be derived.

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A. FAME DATA COMPUTATIONAL METHODS

The following section provides detailed guidance for the data manipulation undertaken in this analysis. I emphasise reproducibility due to the discussion on section 1 ‘Introduction’: this analysis offers a substantive baseline with advanced data science techniques and econometric methods, on top of which I leave it for future analysis to carry out the more sophisticated extensions of section 6. All scripts are publicly available online²⁸, written in Python and Jupyter notebooks.

A.1. DATA CLEANING PROCEDURE

DuckDB operates locally, meaning minimal lag from issuance to execution time provided the operating machine is sufficiently high-compute, and stores data in column vectors, which is a significant advantage to manipulating whole variables. Any SQL commands issued from the Python are compiled down to C, making execution for column transformation operations extremely fast, even for joining tables of several 100s of millions of rows²⁹, as is calculated in this analysis for target-peer spatial distances. I wrote database interaction commands using *Ibis*, an Apache-2.0 licensed Object-Relational Mapping (ORM) framework which allows SQL commands to be written natively in Python³⁰, minimising the likelihood of errors when issuing queries loops or changing variable names. This architecture choice is extremely well-suited for mass data econometric analysis and critical to solving the compute limitations of iterative large- n dataset manipulations required to calculate the valid instruments identified in section 4.1.

Raw data are distributed across a series of Excel files on balance sheet indicators, including time-invariant characteristics for each firm, bucketed into *a1_ID* and *a5_misc* files, and annual data published on a calendar-year basis, bucketed into *a2_key_finance*, *a3_assets*, and *a4_profits* tables.

I traversed the file structure to build a dictionary to the filepaths organised in their subfolders, which are named by the 2-digit industry SIC division, and batch-imported data belonging to the firms in that industry. Some industries have prodigious registered firms, such as SIC division 47 (Retail Trade, 1090 files) and 82 (Office administrative, office support and other business support activities, 1065 files)³¹. I imposed a batch limit of 200 for stable memory handling, splitting up these large industries, and used Python’s asynchronous *concurrent.futures* package to execute the ingestion of each batch, exploiting multithreaded processing. In total, 12,800 files were processed across the 88 available SIC divisions. Time-

²⁸ <https://github.com/alloysiuslip/ucl-diss-ECON0079>

²⁹ Explanatory numbers given in this section are rounded to 3 significant figures for ease

³⁰ The Ibis Project Authors (2026)

³¹ Office for National Statistics (2026d)

invariant firm characteristics were saved to a table named *fixed*, annual firm data to a table named *yearly* within *DuckDB*.

All firm data are tagged with a *registered_number* property corresponding to a Companies House ID. I deduplicated 896,000 *fixed* firm entries using a *registered_number* match, of which the vast majority were exact duplicate entries and the remainder had some mismatch of 1 or more firm characteristics, such as the listed company name, global operating owner, and in rare cases, location. Duplicate entries are an undesirable feature of the Fame dataset, worsened by Fame’s construction from snapshots of reported firm data across timepoints, where resolution of conflicting *fixed* data may be imperfect. Fame also contains data impute errors, meaning variations in names referring to the same entity. When duplicates occurred, I effectively randomly chose an entry to keep. A more robust analysis could devise further data science methods, such a deployment of an acceptable Levenshtein distance to resolve differing name entries, or separately treating firms with multiple listed addresses. Further data cleaning procedures are discussed in Irastorza-Fadrique et al. (2026), notably these include avoiding double-counting of firms through both their parent company and subsidiaries, truncation of data to at earliest 2010 due to changes in reporting standards, and reclassification of firm SIC codes using NLP on firm business activity descriptions.

As the main analysis here only uses location data as the *fixed* characteristic of analysis, due to its time constraints, and due to the low number of duplicates and other cleaning errors listed above relative to the size of the dataset, I treat those errors as random effects and deem the impact of the random selection strategy minimal. I similarly removed duplicate firm-year entries in the *yearly* data, and merged the two tables based on ensuring every *fixed* firm identity had a corresponding *yearly* set of rows, and vice versa. For efficiency, I adopted a ‘max-merge’ column strategy, which is effective for resolving entries which are zero and non-zero in conflicting duplicates. This however means that any two observations with conflicting non-zero values keep the large value. If it is random errors causing Fame to have these conflicting, duplicate values, this strategy has the potential to bias all financial data entries upwards, however this bias is unlikely to be correlated with any of this investigation’s variables of interest. I further removed a small minority (<200) firm entries not based in the UK, which can be identified through their *registered_number* format. In total, deduplication reduced the number of firms from 9,280,000 to 8,220,000, and the number of panel observations from 44,900,000 to 42,900,000.

I restrict the sample to observations with more than 10 employees, and drop any observation from the *yearly* data missing *renumeration_employees* and *ebitda*

properties which are required to calculate firm GVA. This follows Irastorza-Fadrique et al. (2026): low-employee firms, additionally more likely to be shell companies, are likely to mechanically yield an inaccurately high TFP-residual calculation. Doing this at the observation rather than firm, has the risk of inducing bias to the variables measured as there may remain firms in the sample straddling the employees threshold which endogenously have their observation years of low-employees excluded. First-differencing in panel regressions with this data provides some mitigation to this issue by providing a buffer of one year either side to excluded observations. Nevertheless, this analysis adopts the observation-level threshold strategy. This set of filters reduced the number of valid firms in this analysis to 152,000 with 1,130,000 panel observations.

Figure VI shows the distribution of active firms across years, which follows

FIGURE VI
Panel observation availability by year

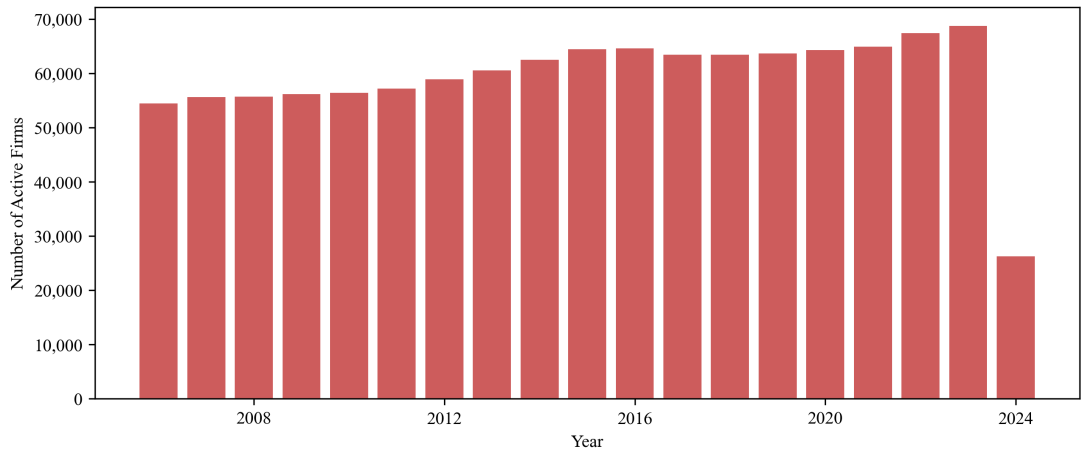
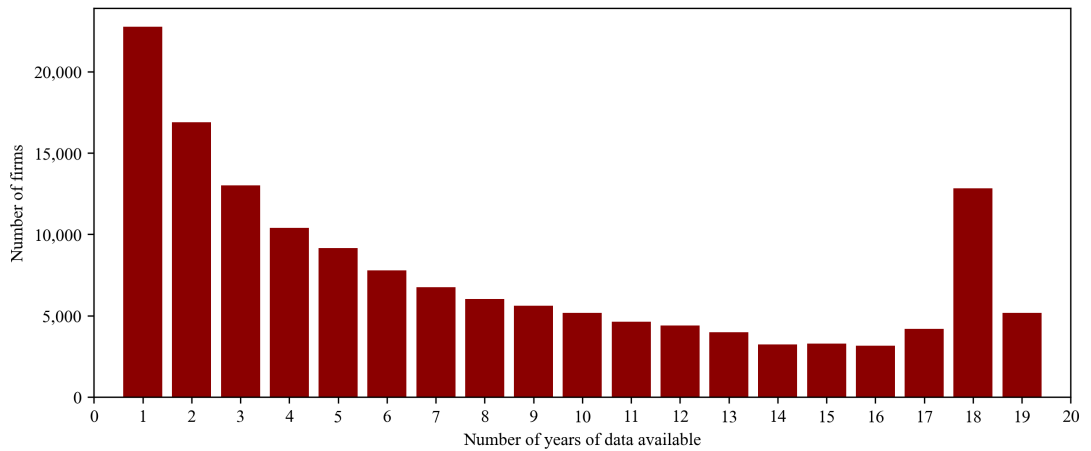


FIGURE VII
Distribution of years of data available per firm



Fame panel sample downloaded from calendar years 2006 to 2024 inclusive. Firms with zero years of data available filtered out. Sample downloaded February 2025, resulting in incomplete reporting for 2024.

expected trends from macroeconomic business population surveys³². We can note an observable dip after the 2016 Brexit referendum and during the COVID-19 pandemic. We can also observe that 2024 is an incomplete sample set as not all firms had submitted their financial accounts for that year, but this is treated as a random effect and no further firms are filtering as a result of this.

The lifespan of a firm, and the corresponding number of *yearly* observations within the dataset follows a geometric distribution as given in Figure VII. With each passing year, a firm has a binary survival-remain decision, with an increasing discrete probability of survival for older firms.

³² for Business and Trade (2025)

B. EXTENSIONS

B.1. SELECTION OF PARAMETERS

In section 3, I recorded quality of location data for each firm, but discussed limitations of applications of that data.

In section 3.5, I discussed arbitrary choices over network and distance restrictions, using *ttwa* boundaries and 5km max distances.

These could be improved by cross-validation.

B.2. IMPROVING ESTIMATES BY GRANULARITY

This approach takes a holistic, mass microdata approach, which has the advantage of testing the limits of the structural model, and searching for baseline patterns of universal and persistent trends across time-firm observations. However, it omits investigation of serious economic consequence to explore hyper-regional, clustering phenomena that are core to the urban development narrative discussed in 2. This has serious risk of trying to impose a single, 3-factor production function specification, and unified parameter estimate in each of δ_0, ξ_0, ζ_0 for all firms, which represented a unified, mass of heterogeneity over their respectively differing industries, sizes, ages, practices, and fixed location and cultural factors. Some of these will be captured in firm fixed effects, leaving a very low R^2 of remaining variation that be captured in this aggregated analysis, other components of it will be time-varying.

The strongest advantage of this investigation’s approach is that it creates a platform for defensible regressions with the rich microdata handled, ready to estimate in new choice specifications of the researcher’s interest. Future analysis should proceed with policy-relevant choice cuts. Section 1 outlined several key and emerging questions governing the economic and policy future of the UK. Evidence supporting these cases for regional policy would greatly benefit from the quantitative, restriction-free analysis which this paper establishes and explores.

Some suggested cuts of data could be:

- Estimating and monitoring the strength of knowledge spillovers in the UK government’s identified ‘innovation’ clusters (E.Table XVI)
- Evaluating the spillover strength of the UK’s high-performing cities (E.Table XVII) and towns (E.Table XVIII)

I begin preliminary estimation and analysis of these phenomena, filtering the corpus of firms to selection on a place-industry basis, and estimating the same δ_0, ξ_0, ζ_0 parameters. Results for a static, distance-decay model are given in

Appendix E. However, similar to results given in the rest of the paper, these results are not significant, nor stable, for instance keeping within the *a priori* limits of $|\theta_0| < 1, \forall \theta_0 \in \boldsymbol{\theta}_0$.

We should therefore conclude from this that mere heterogeneity over the analysis' corpus of firms is not the cause of the parameter instability, the issues discussed above requiring further econometric investigation or data revision are likely necessary to bring the empirical component of this analysis to stable estimates. Nevertheless, this short exercise taking cuts of data demonstrates how policy-focussed investigations to derive credible empirical estimates would proceed from the framework I have established here.

B.3. NETWORK FIXED EFFECTS

Local demand shocks can be thought of as Manski (1993)'s *correlated effects* discussed in section 2.2, such that $\mathbb{E}[\varepsilon_{i,t,g}\varepsilon_{i,t,h}] \neq 0$, where (g, h) are some close members of a network or group within the interactions matrix $\mathbf{W}$. This is a phenomenon specific to peer effects analysis and differs from the usual omitted variable bias (OVB) that impacts all firms discussed in standard econometric theory; here, specific identity of a group or network exposes members to a certain third-party common shock, for which a researcher should introduce controls. Manski (1993) labels these *correlated effects*. Econometrically, a *correlated* shock is some omitted variable that hits both error terms, such as a demand shock in the local area $v_{i,t} \rightarrow \varepsilon_{i,t}, \varepsilon_{j,t}$. This would simultaneously lift the recorded profit and GVA, of both a target and its peers. Another instance could be an exogenous change in the mutual supply chain of a firm and its peer, which is most likely to occur when i, j share an industry. A negative price cost shock, for instance, would lower the cost of inputs for both firms and be accounted for in the short run by increased GVA for the value of inputs indexed using macroeconomic data.

It may be considered an arbitrary choice what is a sufficiently 'local' region to capture these effects and model them as a variable correlated with regressors. Fortunately, econometric theory through Bramoullé et al. (2009), discusses the use of *local* network fixed effects which can be applied to existing $\mathbf{W}$ weighting matrices. Network-fixed effect reduced-form regressions are estimated by taking $(\mathbb{I} - \mathbf{W})$ transformations of all parameters: dependent variable, regressors, and instruments. In my analysis, this means the existing applied weighting matrices $\mathbf{W}_g, \mathbf{W}_d \dots$ for static network and group approaches respectively. A corollary for *global* network fixed effects is the use of a similar transformation $(\mathbb{I} - \mathbf{V})$ which

follows the assignment rule:

$$w_{ij} \in \mathbf{V} = \begin{cases} 0, & \text{if } w_{ij,l} \neq 0, \exists l < k \\ \frac{1}{s_g}, & \text{else if } (i, j) \in g \\ 0, & \text{otherwise} \end{cases}$$

This shares notable similarities with the group matrix $\mathbf{W}_g$: it is a ‘leave-on-in’, average of peers group transformation, demeaned. For application to the group model, $\mathbf{V}$ and $\mathbf{W}_g$ are near-identical, except for the use of s_g instead of s_{g-1} as the denominator for $\mathbf{V}$, and a target’s firm variable of interest is taken as part of the weighted average of its own peers. In extensions on section 3.5, it is n_g is calculated at the *ttwa* level to align with all firms within the mathematical ‘network’ of the set of distance-weighted firms.

As briefly laid out in section 3, identification holds through instruments $(\mathbb{I} - \mathbf{W})\mathbf{W}^2x$ on the endogenous $(\mathbb{I} - \mathbf{W})\mathbf{W}y$ regressor under certain conditions of $\mathbf{W}$. The formal test is linear independence of the $\mathbf{I}, \mathbf{W}, \mathbf{W}^2, \mathbf{W}^3$ and the same limits to computational feasibility apply to this higher-order ($n \times n$) set of matrices evaluation. Nevertheless, Bramoullé et al. (2009) suggests that more than $s_g \geq 3$ distinct group sizes assumed to be exogenous, or $s_d \geq 3$ network diameter should be sufficient conditions to the linear independence of the high-order matrices. The equivalent argument applies for *global* network fixed effects: they overcome the ‘Reflection Problem’ via the instrument $(\mathbb{I} - \mathbf{V})\mathbf{W}^2x$ and its higher-order equivalents.

$$\begin{aligned} (\mathbb{I} - \mathbf{W})\mathbf{y}_t &= \gamma_t(\mathbb{I} - \mathbf{W})\boldsymbol{\iota} + \beta_0 + \beta_1(\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{k}_t + \beta_2(\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{l}_t \\ &+ \beta_3(\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{y}_t + \beta_4(\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{k}_t + \beta_5(\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{l}_t + (\mathbb{I} - \mathbf{W})\boldsymbol{\varepsilon}_t \end{aligned} \quad (47)$$

$$0_{1 \times 2} = \mathbb{E}[\varepsilon_{i,t} \begin{pmatrix} (\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{k}_t & (\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{l}_t \end{pmatrix}]$$

$$\begin{aligned} (\mathbb{I} - \mathbf{V})\mathbf{y}_t &= \gamma_t(\mathbb{I} - \mathbf{V})\boldsymbol{\iota} + \beta_0 + \beta_1(\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{k}_t + \beta_2(\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{l}_t \\ &+ \beta_3(\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{y}_t + \beta_4(\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{k}_t + \beta_5(\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{l}_t + (\mathbb{I} - \mathbf{V})\boldsymbol{\varepsilon}_t \end{aligned} \quad (48)$$

$$0_{1 \times 2} = \mathbb{E}[\varepsilon_{i,t} \begin{pmatrix} (\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{k}_t & (\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{l}_t \end{pmatrix}]$$

This specification omits a firm fixed effect component α_i in equations (47) and (48), and accordingly has no need to first-difference the reduced form equation. This is due to the risk of applying of both firm and network fixed effects simultaneously to be multicollinear. I run the extension models (47) and (48) for

the static group and network weights $\mathbf{W} \in \mathbf{W}_{g1}, \mathbf{W}_{d1}$. Despite efforts with the above, I find that all specifications failed the Sargan-Hansen J test, which implies over-identifying restrictions on instruments such that the GMM-CUE estimator could not converge to the required zero moment conditions. Estimates also suffer from multicollinearity anyway, giving parameter coefficients which fall far out of the expected range of $|\delta_0|, |\xi_0|, |\zeta_0| < 1$. Unstable estimates such as these should not be considered consistent, so I do not report them here.

Further analysis could investigate this issue more thoroughly and successfully implement these network fixed effects as the key to stripping out local demand effects or supply shocks.

B.4. ALTERNATIVE LOCATION DATA

This analysis thoroughly explained the advantages of Fame’s high observation count and the subsequent data science employed to handle its nuances.

Alternative datasets were considered for this investigation. This investigation aimed to incorporate the ‘Longitudinal Business Database’, a data spine designed to be implemented in a similar panel format to presented here with Fame, with consistent estimates over time?. This database is a novel innovation in data collecting procedures employing innovative techniques for consistency of data such as double-snapshotting, offers a similar time period coverage to Fame from 1999-2024, and is comprehensive as a corpus of firm in the UK, again similar to Fame, as it derived from the ONS’ Inter-Departmental Business Register. It offers 2 crucial advantages:

1. It tracks firms at the site (*local unit*) level, solving the multisite and attribution of GVA issue discussed in section 3.
2. Its use of double-snapshotting permits tracking of firm movers, allowing a more sophisticated specification of firm entry and exit, beyond the simplistic assumptions of static movers with controls presented here.

As it employs collection methods as a register, rather than a sample, the Longitudinal Business Database is only available to registered ONS Secure Research Service project users. As my registration was finalised halfway through the 3-month timescale available for this MSc Dissertation project, my use of the LDBR was limited to only realising that ingestion and cleaning methods employed for Fame would require significant revisions for the LBD’s structure, so a planned joint approach was dropped. Nevertheless, the LBD continues to be a viable path for extending this analysis.

B.5. VECTOR TRANSFORMATIONS PROCEDURE

Computation of a matrix product, including taking higher indices, of the interactions between 1,520,000 firms is not computationally difficult, especially with sophisticated algorithms to handle $\mathbf{W}$'s block diagonal structure. However, storing the calculation in temporary memory is beyond the capacity of any standard machine available as a research tool.

Instead, I use a computational approach that avoids calculating the $\mathbf{W}^2, \mathbf{W}^3 \dots$ matrix separately, but rather iteratively applies the last pre-multiplied matrix ($n \times n$) to the aggregate variable ($n \times 1$). This allows repeated use of the transformation matrices $\mathbf{W}_{g1}, \mathbf{W}_{d1}, \dots$ without having to compute a new transformation matrix $\mathbf{W}^2 \dots$. Additionally, the 'native' matrices $\mathbf{W}$ are easier to apply as transformations as their weight values can be computed on-demand using the group-network structure procedures, whereas each entry of an entirely new matrix $\mathbf{W}^2$ would have to be stored in its own table.

Using memory-efficient *DuckDB*, I designed the following procedure:

1. Establish a set of transforms that could be applied many-to-many to a set of vectors.

Ex 1: {variables : y, k, l , transforms : $g1, g2, g3$ }

Ex 2: {variables : $wg1_k, wg1_l$, transforms : $g1$ }

2. Parse this many-to-many relation to a one-to-one relation, that maps onto the new column vector post-transform. This requires a function with a consistent set of rules to apply to a transformation-vector combination. Map each entry in the one-to-one relation to a 'depth' level, corresponding to number of new vectors that need to be created first before the right vector exists to apply the latest transformation.

```
Depth 1 (15): [ 'y -> wg1_y', 'y -> wg2_y', 'y -> wg3_y', ... ]
Depth 2 (24): [ 'wg1_k -> w2g1_k', 'wg1_k -> wg2wg1_k', ... ]
Depth 3 (10): [ 'w2g1_k -> w3g1_k', 'w2g1_l -> w3g1_l', ... ]
Depth 4 (10): [ 'w3g1_k -> w4g1_k', 'w3g1_l -> w4g1_l', ... ]
Depth 5 (4): [ 'w4g1_k -> (i-wg1)w4g1_k', ... ]
```

3. Execute the transformations at each depth level in parallel, calculating the table of vectors after each depth as an input to the new depth set of transformations.

Group $\mathbf{W}_g$: sum the target vector over i firms in the panel database, adopting a 'leave-one-out' approach as necessary by subtracting the

current row. Row-normalise by counting the summed rows and taking it as the quotient denominator.

Group W_d : Pre-calculate the distance transformation measures d_1, d_2, d_3 on the distance table and join the panel table to it twice: once for the target $firm_i$ and once for the peer $firm_j$. Take the specified weight column for the transformation (ex: d_1) and multiply it with the specified variable vector. Row-normalise by summing the transformed-distance (weight) column and using as a denominator.

B.6. $\mathbf{W}$ TRANSFORMATIONS SUMMARY STATISTICS

TABLE X
Summary statistics for $\mathbf{W}_d \mathbf{x}$ and high-order transformations

	y	k	l
<i>base</i>	8.505 (1.603)	9.478 (1.924)	4.525 (1.374)
$\mathbf{W}_{d1}$	8.610 (0.913)	9.602 (1.109)	4.592 (0.755)
$\mathbf{W}_{d2}$	8.594 (1.054)	9.582 (1.283)	4.578 (0.876)
$\mathbf{W}_{d3}$	8.551 (0.602)	9.531 (0.740)	4.550 (0.442)
$\mathbf{W}_{d1}^2$		9.652 (1.025)	4.619 (0.700)
$\mathbf{W}_{d1}^3$		9.671 (0.986)	4.629 (0.673)
$\mathbf{W}_{d1}^4$		9.685 (0.961)	4.638 (0.657)
$\mathbf{W}_{d2}^2$		9.621 (1.239)	4.600 (0.848)
$\mathbf{W}_{d3}^2$		9.546 (0.675)	4.556 (0.392)
$(\mathbf{I} - \mathbf{W}_{d1})$	-0.101 (1.539)	-0.121 (1.823)	-0.064 (1.367)
$(\mathbf{I} - \mathbf{W}_{d1})\mathbf{W}_{d1}$	-0.040 (0.638)	-0.049 (0.754)	-0.028 (0.566)
$(\mathbf{I} - \mathbf{W}_{d1})\mathbf{W}_{d1}^2$		-0.019 (0.632)	-0.010 (0.472)
$(\mathbf{I} - \mathbf{W}_{d1})\mathbf{W}_{d1}^3$		-0.014 (0.586)	-0.009 (0.437)
$(\mathbf{I} - \mathbf{V}_{d1})$	-0.000 (1.577)	-0.000 (1.890)	-0.000 (1.365)
$(\mathbf{I} - \mathbf{V}_{d1})\mathbf{W}_{d1}$	-0.000 (0.869)	-0.000 (1.050)	-0.000 (0.738)
$(\mathbf{I} - \mathbf{V}_{d1})\mathbf{W}_{d1}^2$		-0.000 (0.961)	-0.000 (0.679)
$(\mathbf{I} - \mathbf{V}_{d1})\mathbf{W}_{d1}^3$		-0.000 (0.919)	0.000 (0.651)
$(\mathbf{I} - \mathbf{V}_{d1})\mathbf{W}_{d1}^4$		-0.000 (0.892)	0.000 (0.633)

Simple means of each transformed vector reported, standard deviation below in (parenthesis). A cell entry represents the vector transformation, whether higher order $\mathbf{W}_{d2}^2$ or none at all, *base*, to a log aggregate variable, y_t, k_t, l_t .

TABLE XI
Summary statistics for $\mathbf{W}_g \mathbf{x}$ transformations and high-order
variants

	y	k	l
<i>base</i>	8.475 (1.580)	9.443 (1.900)	4.501 (1.354)
$\mathbf{W}_{g1}$	8.711 (1.120)	9.720 (1.373)	4.634 (0.960)
$\mathbf{W}_{g2}$	8.457 (0.577)	9.422 (0.714)	4.483 (0.399)
$\mathbf{W}_{g3}$	8.462 (0.293)	9.430 (0.365)	4.492 (0.169)
$\mathbf{W}_{g1}^2$		9.720 (1.346)	4.634 (0.936)
$\mathbf{W}_{g2} \mathbf{W}_{g1}$		9.684 (0.738)	4.623 (0.488)
$\mathbf{W}_{g3} \mathbf{W}_{g1}$		9.696 (0.349)	4.634 (0.199)
$\mathbf{W}_{g1} \mathbf{W}_{g2}$		9.513 (0.664)	4.508 (0.369)
$\mathbf{W}_{g2}^2$		9.427 (0.690)	4.488 (0.380)
$\mathbf{W}_{g3} \mathbf{W}_{g2}$		9.409 (0.373)	4.475 (0.162)
$\mathbf{W}_{g1} \mathbf{W}_{g3}$		9.464 (0.344)	4.496 (0.154)
$\mathbf{W}_{g2} \mathbf{W}_{g3}$		9.434 (0.359)	4.493 (0.164)
$\mathbf{W}_{g3}^2$		9.436 (0.356)	4.496 (0.159)
$\mathbf{W}_{g1}^3$		9.720 (1.342)	4.634 (0.932)
$\mathbf{W}_{g1}^4$		9.720 (1.341)	4.634 (0.932)
$(\mathbf{I} - \mathbf{W}_{g1}) \mathbf{W}_{g1}^2$		-0.000 (0.868)	0.000 (0.645)
$(\mathbf{I} - \mathbf{V}_{g1}) \mathbf{W}_{g1}^2$		-0.000 (0.442)	0.000 (0.329)
$(\mathbf{I} - \mathbf{W}_{g1}) \mathbf{W}_{g1}^3$		0.000 (0.854)	0.000 (0.634)
$(\mathbf{I} - \mathbf{V}_{g1}) \mathbf{W}_{g1}^3$		0.000 (0.429)	0.000 (0.318)
$(\mathbf{I} - \mathbf{W}_{g1})$	-0.000 (1.501)	0.000 (1.765)	-0.000 (1.389)
$(\mathbf{I} - \mathbf{V}_{g1})$	-0.000 (0.986)	0.000 (1.152)	-0.000 (0.919)
$(\mathbf{I} - \mathbf{W}_{g1}) \mathbf{W}_{g1}$	-0.000 (0.783)	0.000 (0.943)	-0.000 (0.709)
$(\mathbf{I} - \mathbf{V}_{g1}) \mathbf{W}_{g1}$	-0.000 (0.432)	0.000 (0.517)	-0.000 (0.392)
$(\mathbf{I} - \mathbf{W}_{g1}) \mathbf{W}_{g1}^4$		-0.000 (0.851)	-0.000 (0.631)
$(\mathbf{I} - \mathbf{V}_{g1}) \mathbf{W}_{g1}^4$		-0.000 (0.426)	-0.000 (0.316)

C. STATIC GROUPS: ADDITIONAL RESULTS

TABLE XII
First-stage 2SLS (panel OLS) results

	(1.) lag2	(2.) lag3	(3.) lag4
const	4.152 (0.046)	1.896 (0.030)	2.330 (0.028)
$\mathbf{W}_{g1}^2 \mathbf{k}$	0.263 (0.005)	0.048 (0.005)	-0.660 (0.026)
$\mathbf{W}_{g1}^2 \mathbf{l}$	0.433 (0.007)	0.088 (0.005)	-1.132 (0.029)
$\mathbf{W}_{g1}^3 \mathbf{k}$		0.324 (0.006)	0.327 (0.006)
$\mathbf{W}_{g1}^3 \mathbf{l}$		0.603 (0.006)	0.614 (0.006)
$\mathbf{W}_{g1}^4 \mathbf{k}$			0.679 (0.025)
$\mathbf{W}_{g1}^4 \mathbf{l}$			1.169 (0.028)
i F.E.	✓	✓	✓
t F.E.	✓	✓	✓
Obs.	726,283	726,283	726,283
R ² -adj (excl. F.E.)	0.2178	0.5632	0.5887
R ² -adj (Overall)	0.4646	0.7812	0.8104

TABLE XIII
Static panel IV, 3 orthogonal groups

	(1.) exog-2	(2.) iv-2	(3.) exog-3	(4.) iv-3
β_1	0.261 (0.005)***	0.233 (0.008)***	0.261 (0.005)***	0.236 (0.011)***
β_2	0.650 (0.005)***	0.555 (0.008)***	0.649 (0.005)***	0.580 (0.020)***
δ_1	0.198 (0.004)***	-0.337 (0.191)*	0.198 (0.004)***	-0.715 (0.596)
δ_2	0.020 (0.006)***	2.812 (1.038)***	0.019 (0.006)***	4.312 (3.073)
ξ_1	-0.006 (0.002)**	0.025 (0.015)*	-0.006 (0.002)**	0.047 (0.043)
ζ_1	0.011 (0.003)***	0.037 (0.015)**	0.011 (0.003)***	0.048 (0.030)
ξ_2	-0.007 (0.004)*	-0.393 (0.143)***	-0.007 (0.004)*	-0.477 (0.375)
ζ_2	0.011 (0.006)*	-0.203 (0.088)**	0.011 (0.006)*	-0.331 (0.179)*
δ_3			0.048 (0.017)***	-40.411 (25.266)
ξ_3			-0.009 (0.010)	5.323 (3.293)
ζ_3			0.019 (0.016)	2.586 (1.100)**
J-test		✓ (0.017)**		X (0.559)
Instr.		$\mathbf{W}_{g1}^2 \mathbf{k}, \mathbf{W}_{g1}^2 \mathbf{l}$ $\mathbf{W}_{g2}^2 \mathbf{k}, \mathbf{W}_{g2}^2 \mathbf{l}$		$\mathbf{W}_{g1}^2 \mathbf{k}, \mathbf{W}_{g1}^2 \mathbf{l}$ $\mathbf{W}_{g2}^2 \mathbf{k}, \mathbf{W}_{g2}^2 \mathbf{l}$ $\mathbf{W}_{g3}^2 \mathbf{k}, \mathbf{W}_{g3}^2 \mathbf{l}$
i F.E.	✓	✓	✓	✓
t F.E.	✓	✓	✓	✓
Obs.	719,403	581,797	718,743	581,185
R ² (excl. F.E.)	0.3543	-0.4604	0.3543	-11.5040
R ² (Overall)	0.7967	0.0000	0.7970	0.0000

D. STATIC DISTANCE-DECAY: ADDITIONAL RESULTS

TABLE XIV
First-stage 2SLS (panel OLS) results

	(1.) lag2	(2.) lag3	(3.) lag4	(4.) lag2-noz	(5.) lag3-noz	(6.) lag4-noz
const	4.099 (0.045)	1.691 (0.025)	1.805 (0.024)	3.913 (0.052)	1.568 (0.033)	1.685 (0.034)
$\mathbf{W}_{d1}^2 \mathbf{k}$	0.277 (0.005)	0.024 (0.004)	0.355 (0.013)	0.312 (0.007)	−0.016 (0.008)	0.185 (0.014)
$\mathbf{W}_{d1}^2 \mathbf{l}$	0.397 (0.007)	0.025 (0.005)	0.144 (0.018)	0.361 (0.009)	−0.064 (0.010)	0.046 (0.018)
$\mathbf{W}_{d1}^3 \mathbf{k}$		0.373 (0.005)	0.392 (0.005)		0.440 (0.011)	0.470 (0.012)
$\mathbf{W}_{d1}^3 \mathbf{l}$		0.640 (0.006)	0.646 (0.006)		0.707 (0.013)	0.722 (0.014)
$\mathbf{W}_{d1}^4 \mathbf{k}$			−0.361 (0.014)			−0.239 (0.018)
$\mathbf{W}_{d1}^4 \mathbf{l}$			−0.124 (0.020)			−0.132 (0.025)
<i>i</i> F.E.	✓	✓	✓	✓	✓	✓
<i>t</i> F.E.	✓	✓	✓	✓	✓	✓
Obs.	1,031,359	1,031,359	1,031,359	1,025,318	1,025,318	1,025,318
R ² -adj (excl. F.E.)	0.2190	0.5901	0.5964	0.1835	0.5280	0.5329
R ² -adj (Overall)	0.4532	0.7551	0.7639	0.4105	0.7239	0.7322

TABLE XV
Static panel IV using continuous distance-decay $\mathbf{W}_d$, zero
distances excluded

	(6.) dd1-noz	(7.) dd2-noz	(8.) dd3-noz	(9.) dd1-deep3-noz
β_1	0.273 (0.007)***	0.273 (0.007)***	0.270 (0.007)***	0.273 (0.007)***
β_2	0.509 (0.006)***	0.510 (0.006)***	0.516 (0.007)***	0.509 (0.006)***
δ_0	0.776 (0.487)	0.463 (0.203)**	-4.083 (2.512)	1.058 (0.363)***
ξ_0	-0.080 (0.051)	-0.040 (0.018)**	0.450 (0.281)	-0.109 (0.038)***
ζ_0	-0.085 (0.060)	-0.054 (0.026)**	0.480 (0.285)*	-0.119 (0.045)***
J-test	X (0.381)	X (0.609)	X (0.074)*	X (0.672)
Instr.	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$	$\mathbf{W}_{d2}^2 \mathbf{k}, \mathbf{W}_{d2}^2 \mathbf{l}$	$\mathbf{W}_{d3}^2 \mathbf{k}, \mathbf{W}_{d3}^2 \mathbf{l}$	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$ $\mathbf{W}_{d1}^3 \mathbf{k}, \mathbf{W}_{d1}^3 \mathbf{l}$
Obs.	848,718	848,718	848,718	848,718
R ²	0.0712	0.0754	-1.5298	0.0269

E. DATA CUTS: ESTIMATIONS WITH SUBSETS

TABLE XVI
Spillover estimation for 4 select innovation
clusters

	(1.)	(2.)	(3.)
	gtriangle-ls	ldn-creative	3quad-profs
β_1	0.383 (0.143)***	0.322 (0.039)***	0.443 (0.072)***
β_2	0.538 (0.183)***	0.496 (0.033)***	0.472 (0.085)***
δ_0	3.161 (1.801)*	0.630 (0.421)	-0.315 (0.230)
ξ_0	0.372 (0.658)	-0.007 (0.030)	-0.023 (0.040)
ζ_0	-0.384 (0.844)	-0.091 (0.064)	0.047 (0.054)
J-test	X (0.314)	X (0.458)	X (0.735)
Instr.	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$ $\mathbf{W}_{d1}^3 \mathbf{k}, \mathbf{W}_{d1}^3 \mathbf{l}$	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$ $\mathbf{W}_{d1}^3 \mathbf{k}, \mathbf{W}_{d1}^3 \mathbf{l}$	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$ $\mathbf{W}_{d1}^3 \mathbf{k}, \mathbf{W}_{d1}^3 \mathbf{l}$
i F.E.	✓	✓	✓
t F.E.	✓	✓	✓
Obs.	2,091	18,554	4,830
R ²	-2.1875	0.1146	0.0772

TABLE XVII
Heterogenous spillover effects among service-sector firms in UK cities

	(1.)	(2.)	(3.)	(4.)	(5.)	(6.)
	london	slough	sund	leeds	manc	bham
β_1	0.289 (0.014)***	0.348 (0.030)***	0.197 (0.038)***	0.482 (0.073)***	0.262 (0.069)***	0.253 (0.035)***
β_2	0.482 (0.013)***	0.521 (0.036)***	0.528 (0.085)***	0.403 (0.046)***	0.457 (0.096)***	0.502 (0.031)***
δ_0	-0.096 (0.144)	-0.519 (0.309)*	-0.608 (0.299)**	0.133 (0.181)	-13.877 (7.620)*	0.609 (0.768)
ξ_0	-0.008 (0.013)	0.027 (0.023)	-0.044 (0.031)	-0.005 (0.038)	0.990 (1.201)	-0.052 (0.088)
ζ_0	0.034 (0.019)*	0.016 (0.030)	0.242 (0.119)**	0.009 (0.053)	2.280 (1.839)	-0.062 (0.108)
J-test	✓ (0.003)***	X (0.502)	X (0.102)	X (0.337)	X (0.122)	X (0.757)
Obs.	164,278	19,161	1,885	10,806	28,251	17,585
R ²	0.1091	-0.0065	-0.0296	0.1278	-55.2450	0.0764

TABLE XVIII
Heterogenous spillover effects among service-sector firms in UK towns

	(1.) reading	(2.) bsgstoke	(3.) swindon	(4.) milton-keynes	(5.) guild-ald	(6.) worthing
β_1	0.250 (0.066)***	0.428 (0.148)***	0.347 (0.134)***	0.414 (0.036)***	0.303 (0.039)***	0.401 (0.088)***
β_2	0.570 (0.053)***	0.558 (0.069)***	0.347 (0.079)***	0.455 (0.056)***	0.522 (0.053)***	0.400 (0.167)**
δ_0	1.915 (1.000)*	-0.140 (0.338)	-0.240 (0.325)	0.180 (0.219)	-0.026 (0.347)	-0.277 (0.289)
ξ_0	-0.093 (0.154)	0.016 (0.073)	-0.040 (0.043)	0.037 (0.022)*	0.015 (0.051)	-0.012 (0.041)
ζ_0	-0.108 (0.113)	-0.029 (0.055)	0.072 (0.135)	-0.037 (0.039)	0.002 (0.025)	0.032 (0.117)
J-test	X (0.217)	X (0.596)	X (0.113)	X (0.393)	X (0.965)	X (0.712)
Obs.	9,350	2,727	2,830	5,502	8,426	1,069
R ²	-0.9435	0.0950	-0.0003	0.1776	0.0996	0.0675